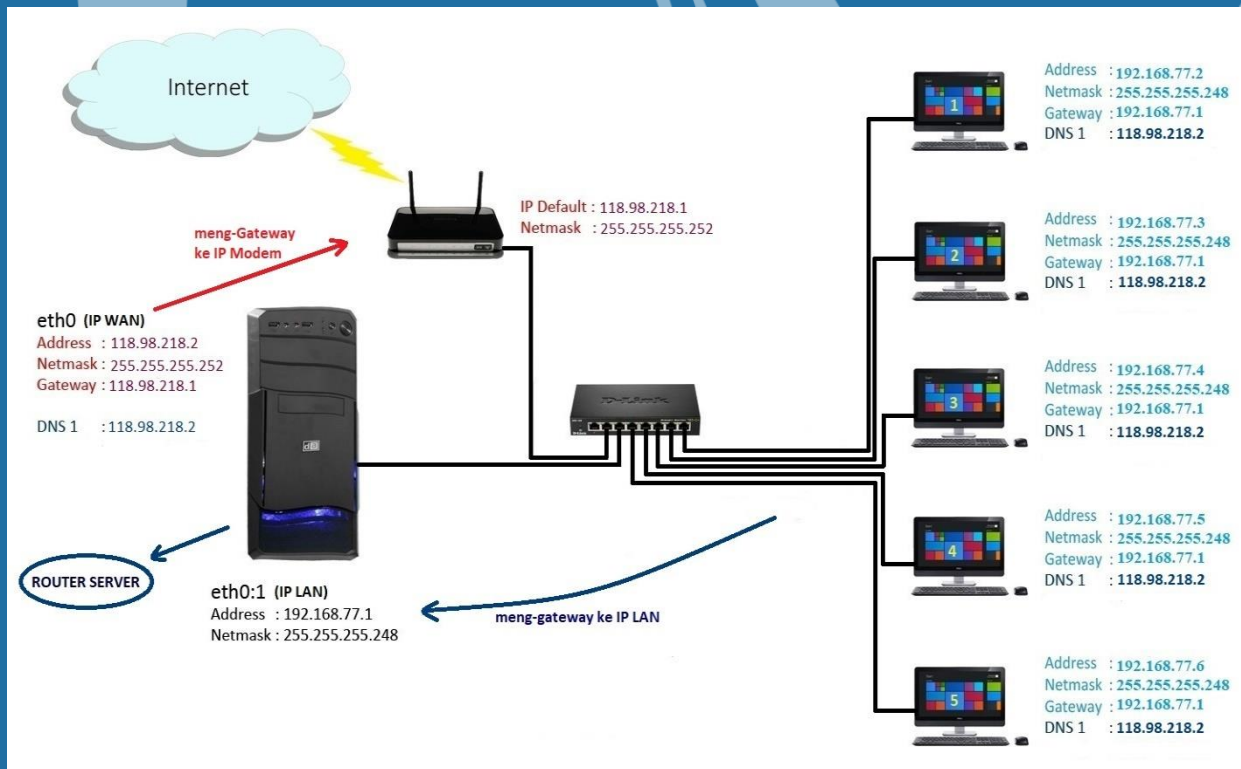


ROUTER SERVER



SALAM SUKSES TKJ CLUB

<http://www.tkjclub.net/tkj>



debian 10

Created by : M. Asriadi

A. Definisi Routing

Routing adalah tehnik atau cara yang di gunakan untuk menghubungkan 2 segmen ip atau lebih yang berbeda, maupun berbeda network.

B. Jenis-Jenis Konfigurasi Routing

Secara garis besar, jenis konfigurasi routing di bagi menjadi 3 jenis, yaitu :

- Routing Default
- Routing dynamic
- Routing Static

1. Routing Default

Routing Default adalah routing yang di gunakan untuk menghubungkan perangkat atau komputer secara manual dengan sistem gateway. Biasanya digunakan pada jaringan yang hanya memiliki 1 jalur keluar atau dengan kata lain hanya untuk jaringan lokal.

2. Routing Dynamic

Routing Dynamic adalah routing yang membuat tabel routing secara otomatis hanya dengan menambahkan IP Network pada router, maka secara otomatis akan mencari jalur atau rute terbaik sesuai dengan konfigurasi yang di buat.

3. Routing Static

Routing Static adalah jenis konfigurasi routing itu sendiri yang settingannya terdapat tabel static routing yang di buat oleh administrator jaringan.

C. Definisi Firewall

Firewall adalah suatu sistem keamanan jaringan yang mengatur atau mengontrol keluar masuknya suatu informasi dari jaringan lokal ke jaringan internet begitupun sebaliknya.

Sejak pertengahan 90-an Linux telah disisipkan fitur baru, yaitu adanya firewall. Teknologi ini telah mengalami perubahan beberapa kali (generasi sebelumnya dikenal dengan ipfwadm atau ipchains) dan sekarang yang kita kenal sebagai packet filtering yaitu iptables.

Iptables adalah aplikasi yang ada pada sistem operasi linux yang berfungsi melakukan filter terhadap trafik pada jaringan. Jadi firewall yang ada pada sistem operasi Linux ini dijalankan oleh iptables.

Selain menjalankan sebagai firewall, iptables juga berfungsi menjalankan fungsi NAT. NAT itu sendiri merupakan internet gateway yang menghubungkan jaringan lokal dengan jaringan internet.

D. Script

```
# ip a l
```

```
# nano /etc/default/grub
```

>>> Cari script **GRUB_CMDLINE_LINUX=""** lalu edit menjadi :

```
GRUB_CMDLINE_LINUX="net.ifnames=0 biosdevname=0"
```

Save dengan Ctrl + X >> tekan Y >> tekan Enter

```
# grub-mkconfig -o /boot/grub/grub.cfg
```

```
# reboot
```

```
# ip a l
```

```
# nano /etc/network/interfaces
```

>>>Ke baris paling bawah, lalu tambahkan script :

```
auto eth0
iface eth0 inet static
    address IP WAN
    netmask Netmask
    gateway IP Modem

auto eth0:1
iface eth0:1 inet static
    address IP LAN
    netmask Netmask
```

Save dengan Ctrl + X >> tekan Y >> tekan Enter

```
# /etc/init.d/networking restart
```

```
# ping IP WAN
```

```
# ping IP LAN
```

```
# nano /etc/resolv.conf
```

>>> Tambahkan script berikut :

```
nameserver DNS 1
```

Save dengan Ctrl + X >> tekan Y >> tekan Enter

```
# nano /etc/init.d/routing
```

```
#!/bin/bash
echo "1" > /proc/sys/net/ipv4/ip_forward
iptables -F
iptables -t nat -F
iptables -P INPUT DROP
iptables -A INPUT -i lo -j ACCEPT
iptables -A INPUT -i eth0 -j ACCEPT
iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT
iptables -t nat -A POSTROUTING -s IP LAN/prefix -o eth0 -j SNAT --to IPWAN
iptables -t nat -nvl
```

Save dengan Ctrl + X >> tekan Y >> tekan Enter

```
# chmod 755 /etc/init.d/routing
```

```
# /etc/init.d/routing restart
```

```
# nano /etc/rc.local
```

>>>Tambah script :

```
#!/bin/sh -e
/etc/init.d/routing start
exit 0
```

Save dengan Ctrl + X >> tekan Y >> tekan Enter

```
# chmod 755 /etc/rc.local
```

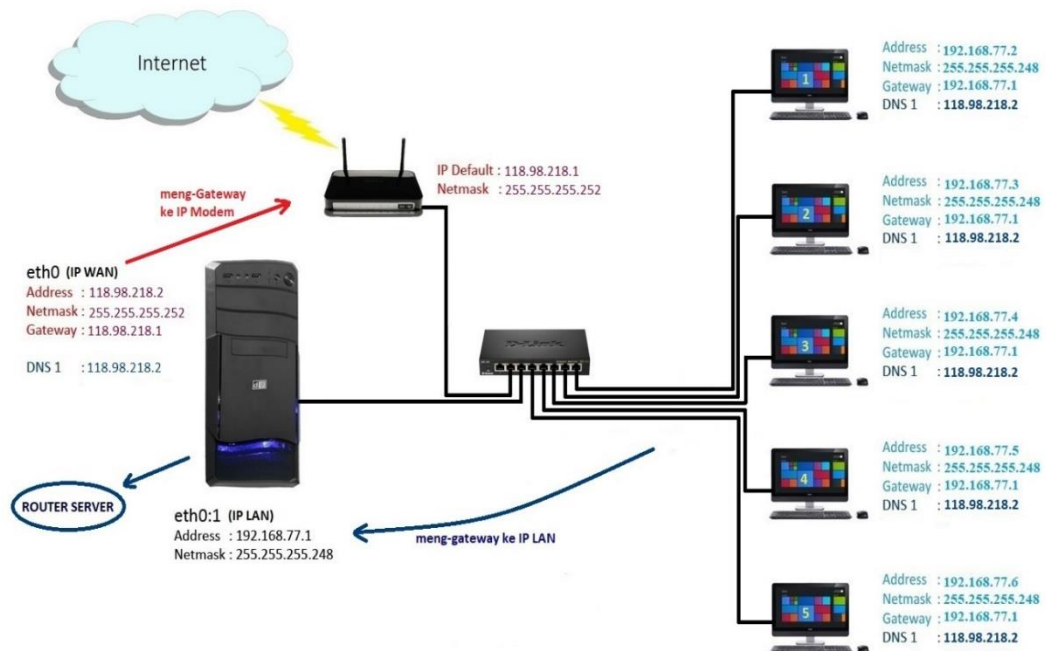
```
# /etc/rc.local
```

Daftarkan IP di Client

```
# ping IP WAN
# ping IP LAN
# ping IP Client }
```

Di CMD

E. Topologi



F. Langkah-Langkah Konfigurasi Router Server

1. Pastikan sudah melakukan penginstalan Linux Debian 10.5 pada virtual mesin yang di buat. Kemudian login user dan password menggunakan super user yaitu **root**.
2. Setelah login maka tampilannya seperti gambar berikut. Setelah itu cek terlebih dahulu nama interfaces-nya, ketikkan perintah berikut :

ip a l >> lalu tekan Enter

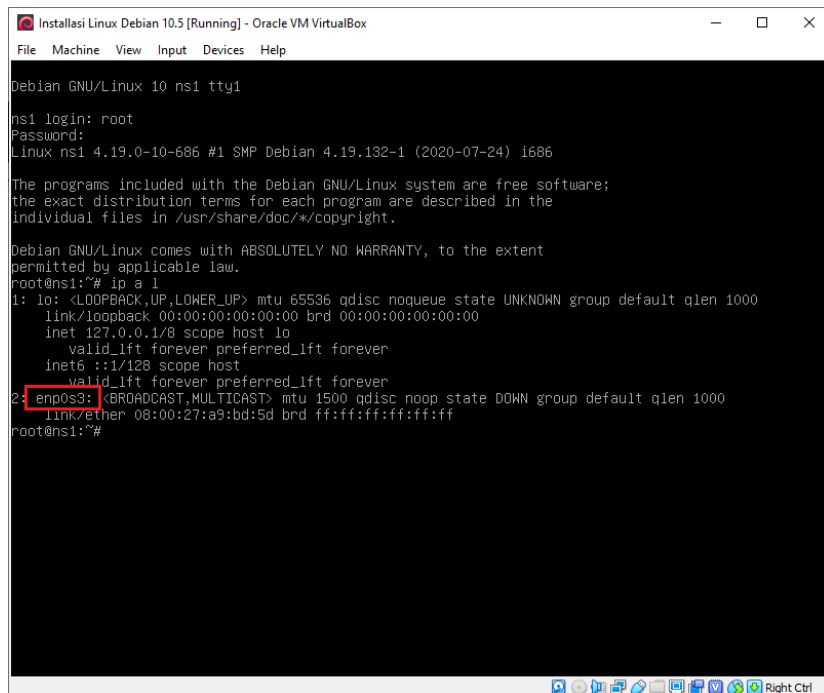
```
Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Debian GNU/Linux 10 ns1 tty1
ns1 login: root
Password:
Linux ns1 4.19.0-10-686 #1 SMP Debian 4.19.132-1 (2020-07-24) i686

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@ns1:~# ip a l
```

3. Secara default nama interfaces pada Linux Debian 10.5 menggunakan “enp0s3” seperti pada gambar dibawah:



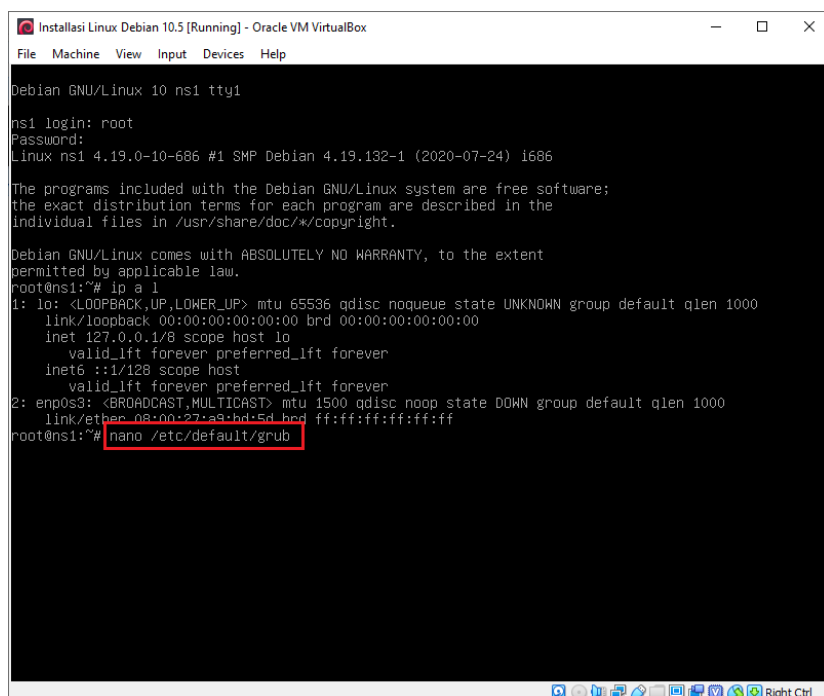
```
Debian GNU/Linux 10 ns1 tty1
ns1 login: root
Password:
Linux ns1 4.19.0-10-686 #1 SMP Debian 4.19.132-1 (2020-07-24) i686

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@ns1:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:a9:bd:5d brd ff:ff:ff:ff:ff:ff
root@ns1:~#
```

4. Lakukan perubahan nama interfaces dari “enp0s3” menjadi “eth0”. Sebenarnya perubahan ini bisa tidak di lakukan, namun untuk mempermudah mengingat nama interface-nya maka dilakukan perubahan. Ketikkan perintah berikut :

nano /etc/default/grub >> lalu tekan Enter



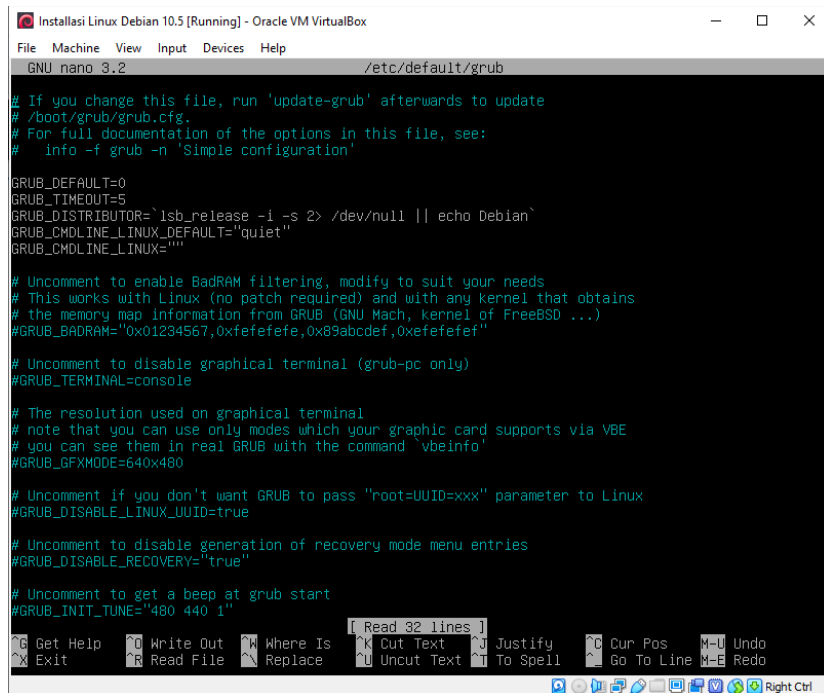
```
Debian GNU/Linux 10 ns1 tty1
ns1 login: root
Password:
Linux ns1 4.19.0-10-686 #1 SMP Debian 4.19.132-1 (2020-07-24) i686

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

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permitted by applicable law.
root@ns1:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:a9:bd:5d brd ff:ff:ff:ff:ff:ff
root@ns1:~# nano /etc/default/grub
```

5. Pastikan tampil seperti gambar di bawah. Kemudian cari script berikut :

GRUB_CMDLINE_LINUX=""

A screenshot of a terminal window titled 'Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox'. The terminal shows the GNU nano 3.2 editor editing the file /etc/default/grub. The content of the file is as follows:

```
# If you change this file, run 'update-grub' afterwards to update
# /boot/grub/grub.cfg.
# For full documentation of the options in this file, see:
#   info -f grub -n 'Simple configuration'

GRUB_DEFAULT=0
GRUB_TIMEOUT=5
GRUB_DISTRIBUTOR=`lsb_release -i -s 2> /dev/null || echo Debian`
GRUB_CMDLINE_LINUX="quiet"
GRUB_CMDLINE_LINUX=""

# Uncomment to enable BadRAM filtering, modify to suit your needs
# This works with Linux (no patch required) and with any kernel that obtains
# the memory map information from GRUB (GNU Mach, kernel of FreeBSD ...)
#GRUB_BADRAM="0x01234567,0xfefefefe,0x89abcdef,0xefefefef"

# Uncomment to disable graphical terminal (grub-pc only)
#GRUB_TERMINAL=console

# The resolution used on graphical terminal
# note that you can use only modes which your graphic card supports via VBE
# you can see them in real GRUB with the command `vbeinfo`
#GRUB_GFXMODE=640x480

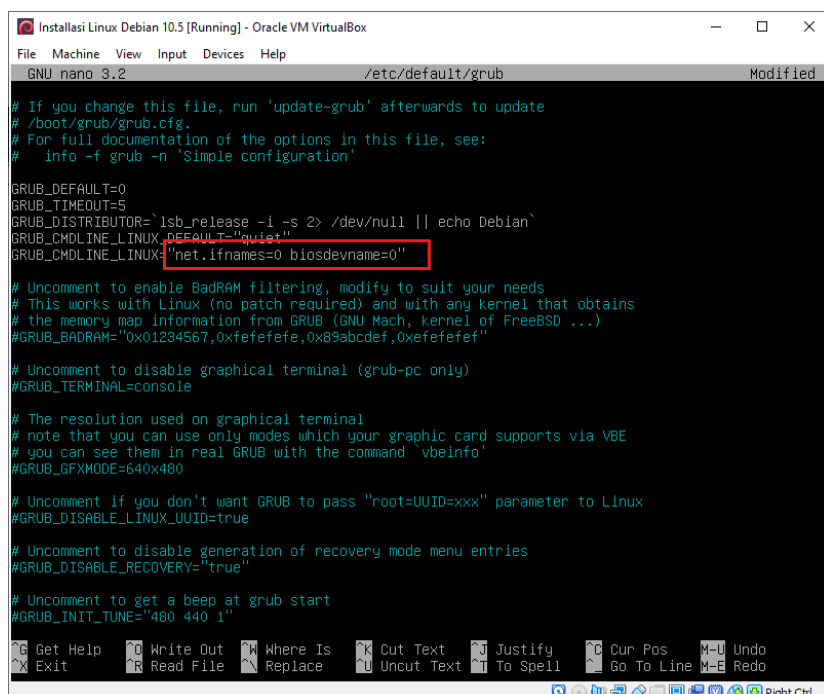
# Uncomment if you don't want GRUB to pass "root=UUID=xxx" parameter to Linux
#GRUB_DISABLE_LINUX_UUID=true

# Uncomment to disable generation of recovery mode menu entries
#GRUB_DISABLE_RECOVERY="true"

# Uncomment to get a beep at grub start
#GRUB_INIT_TUNE="480 440 1"
```

6. Kemudian edit script tersebut menjadi :

GRUB_CMDLINE_LINUX="net.ifnames=0 biosdevname=0"

A screenshot of the same terminal window showing the GRUB configuration file after editing. The line `GRUB_CMDLINE_LINUX="net.ifnames=0 biosdevname=0"` has been added, and the previous line `GRUB_CMDLINE_LINUX=""` has been removed. The word 'Modified' appears in the top right corner of the nano editor window.

```
# If you change this file, run 'update-grub' afterwards to update
# /boot/grub/grub.cfg.
# For full documentation of the options in this file, see:
#   info -f grub -n 'Simple configuration'

GRUB_DEFAULT=0
GRUB_TIMEOUT=5
GRUB_DISTRIBUTOR=`lsb_release -i -s 2> /dev/null || echo Debian`
GRUB_CMDLINE_LINUX="net.ifnames=0 biosdevname=0"

# Uncomment to enable BadRAM filtering, modify to suit your needs
# This works with Linux (no patch required) and with any kernel that obtains
# the memory map information from GRUB (GNU Mach, kernel of FreeBSD ...)
#GRUB_BADRAM="0x01234567,0xfefefefe,0x89abcdef,0xefefefef"

# Uncomment to disable graphical terminal (grub-pc only)
#GRUB_TERMINAL=console

# The resolution used on graphical terminal
# note that you can use only modes which your graphic card supports via VBE
# you can see them in real GRUB with the command `vbeinfo`
#GRUB_GFXMODE=640x480

# Uncomment if you don't want GRUB to pass "root=UUID=xxx" parameter to Linux
#GRUB_DISABLE_LINUX_UUID=true

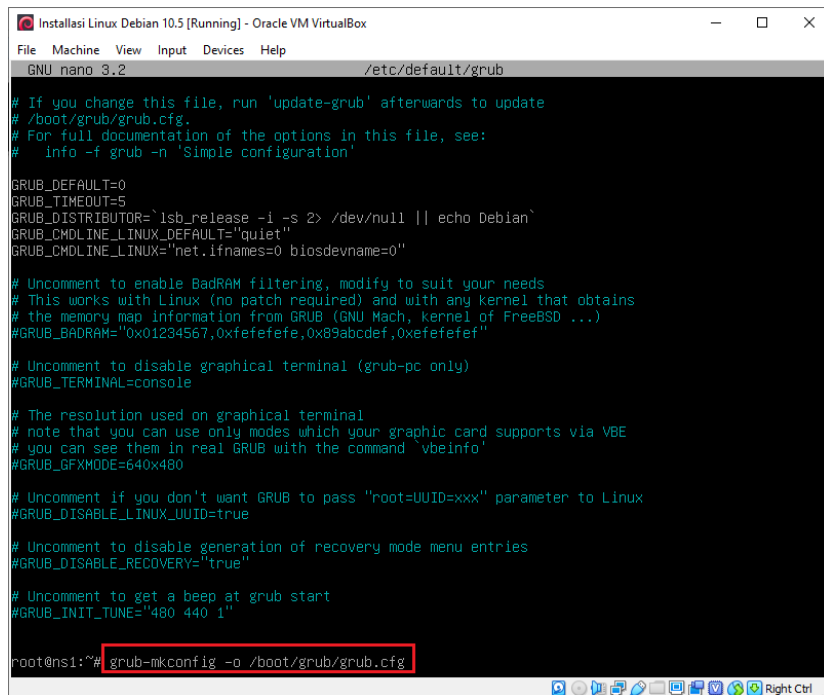
# Uncomment to disable generation of recovery mode menu entries
#GRUB_DISABLE_RECOVERY="true"

# Uncomment to get a beep at grub start
#GRUB_INIT_TUNE="480 440 1"
```

7. Lalu save dengan cara menekan tombol kombinasi **Ctrl + X** lalu tekan tombol **Y** kemudian **Enter**.

8. Setelah itu, buat konfigurasi Grub Bootnya agar interface-nya tidak berubah apabila di restart. Ketik perintah berikut :

grub-mkconfig -o /boot/grub/grub.cfg >> lalu tekan Enter



```
GNU nano 3.2 /etc/default/grub

# If you change this file, run 'update-grub' afterwards to update
# /boot/grub/grub.cfg.
# For full documentation of the options in this file, see:
#   info -f grub -n 'Simple configuration'

GRUB_DEFAULT=0
GRUB_TIMEOUT=5
GRUB_DISTRIBUTOR=`lsb_release -i -s 2> /dev/null || echo Debian`
GRUB_CMDLINE_LINUX_DEFAULT="quiet"
GRUB_CMDLINE_LINUX="net.ifnames=0 biosdevname=0"

# Uncomment to enable BadRAM filtering, modify to suit your needs
# This works with Linux (no patch required) and with any kernel that obtains
# the memory map information from GRUB (GNU Mach, kernel of FreeBSD ...)
#GRUB_BADRAM="0x01234567,0xfefefefe,0x89abcdef,0xefefefef"

# Uncomment to disable graphical terminal (grub-pc only)
#GRUB_TERMINAL=console

# The resolution used on graphical terminal
# note that you can use only modes which your graphic card supports via VBE
# you can see them in real GRUB with the command `vbeinfo`
#GRUB_GFXMODE=640x480

# Uncomment if you don't want GRUB to pass "root=UUID=xxx" parameter to Linux
#GRUB_DISABLE_LINUX_UUID=true

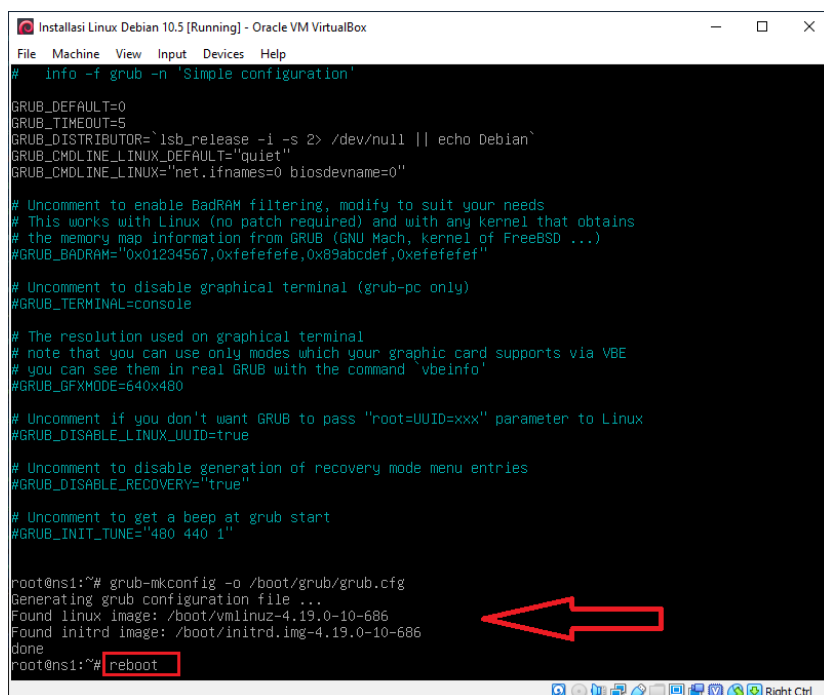
# Uncomment to disable generation of recovery mode menu entries
#GRUB_DISABLE_RECOVERY="true"

# Uncomment to get a beep at grub start
#GRUB_INIT_TUNE="480 440 1"

root@ns1:~# grub-mkconfig -o /boot/grub/grub.cfg
```

9. Hasilnya akan seperti gambar dibawah. Setelah itu, restart server agar konfigurasi yang tadi berjalan dengan baik. Ketikkan perintah berikut :

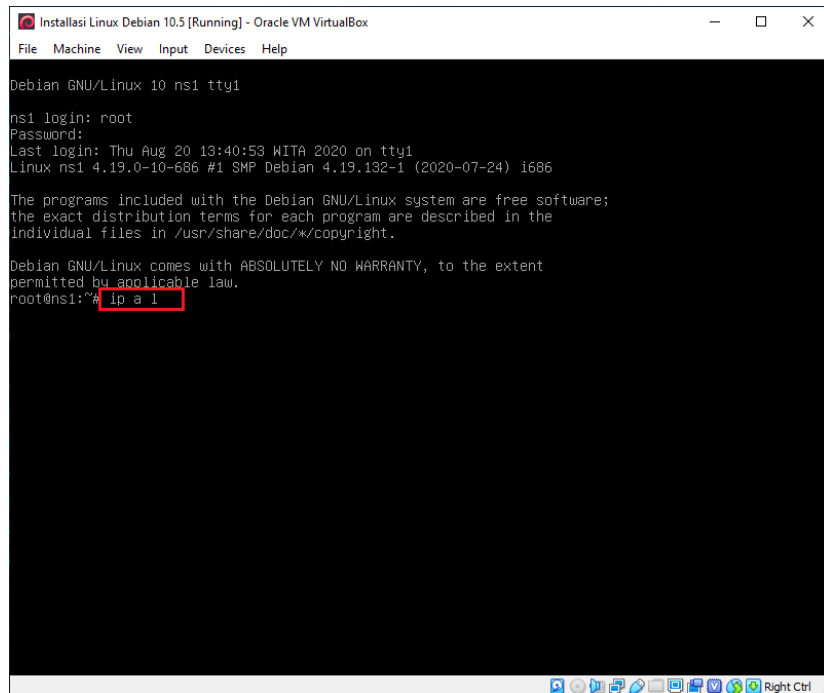
reboot >> lalu tekan Enter



```
root@ns1:~# grub-mkconfig -o /boot/grub/grub.cfg
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-4.19.0-10-686
Found initrd image: /boot/initrd.img-4.19.0-10-686
done
root@ns1:~# reboot
```


10. Kemudian login lagi menggunakan user administrator dan masukkan passwordnya. Setelah itu, cek lagi nama interfaces-nya. Masukkan perintah :

ip a l >> lalu tekan Enter

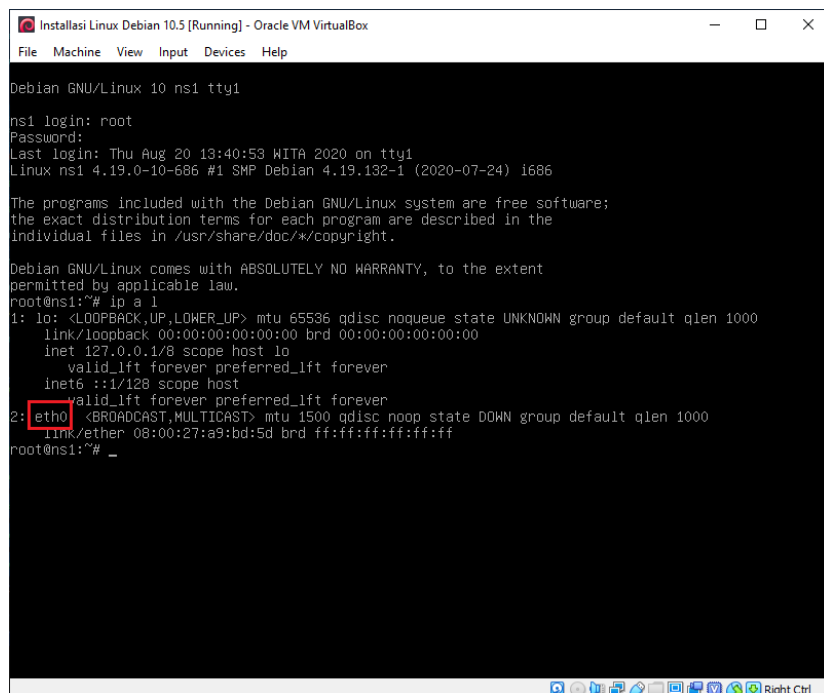


```
Debian GNU/Linux 10 ns1 tty1
ns1 login: root
Password:
Last login: Thu Aug 20 13:40:53 WITA 2020 on tty1
Linux ns1 4.19.0-10-686 #1 SMP Debian 4.19.132-1 (2020-07-24) i686

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individual files in /usr/share/doc/*/copyright.

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permitted by applicable law.
root@ns1:~# ip a l
```

11. Pastikan nama interface-nya sudah berubah dari **enp0s3** menjadi **eth0** seperti gambar di bawah.



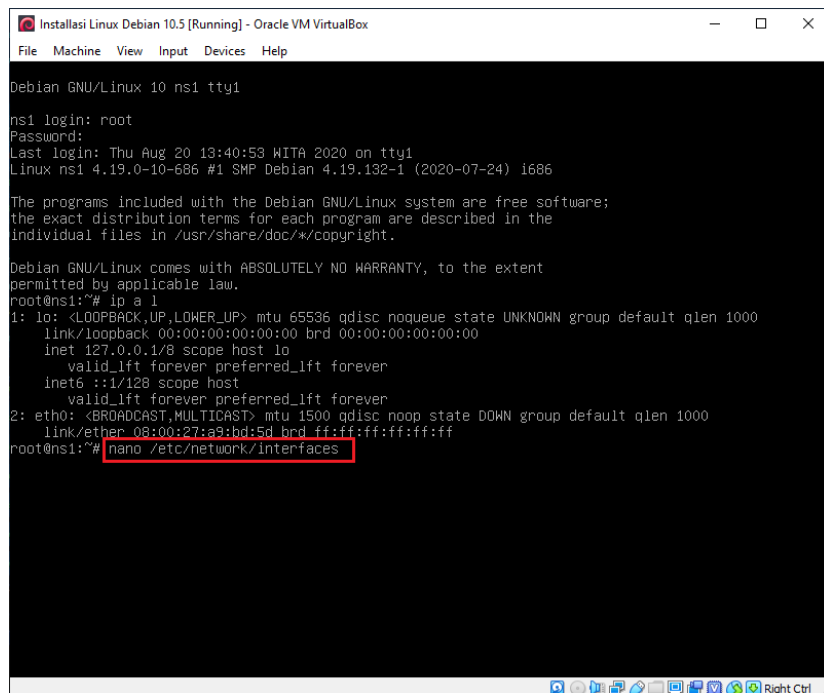
```
Debian GNU/Linux 10 ns1 tty1
ns1 login: root
Password:
Last login: Thu Aug 20 13:40:53 WITA 2020 on tty1
Linux ns1 4.19.0-10-686 #1 SMP Debian 4.19.132-1 (2020-07-24) i686

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individual files in /usr/share/doc/*/copyright.

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permitted by applicable law.
root@ns1:~# ip a l
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:a9:bd:5d brd ff:ff:ff:ff:ff:ff
root@ns1:~#
```

12. Lakukan pendaftaran IP sesuai dengan topologi di atas. Ketikkan perintah berikut :

nano /etc/network/interfaces >> lalu tekan Enter



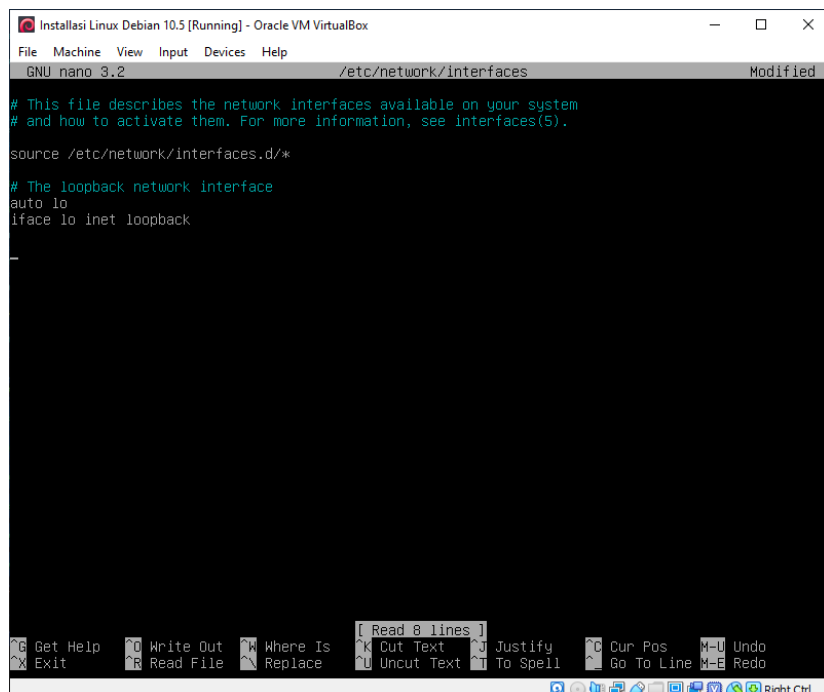
```
Installasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Debian GNU/Linux 10 ns1 tty1
ns1 login: root
Password:
Last login: Thu Aug 20 13:40:53 WITA 2020 on tty1
Linux ns1 4.19.0-10-686 #1 SMP Debian 4.19.132-1 (2020-07-24) i686

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Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@ns1:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:a9:bd:5d brd ff:ff:ff:ff:ff:ff
root@ns1:~# nano /etc/network/interfaces
```

13. Akan tampil seperti gambar di bawah.



```
Installasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
GNU nano 3.2 /etc/network/interfaces Modified

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

[ Read 8 lines ]
Get Help Write Out Where Is Cut Text Justify Cur Pos M-U Undo
Exit Read File Replace Uncut Text To Spell Go To Line M-E Redo
```

14. Tambahkan script berikut di bagian paling bawah :

```
auto eth0

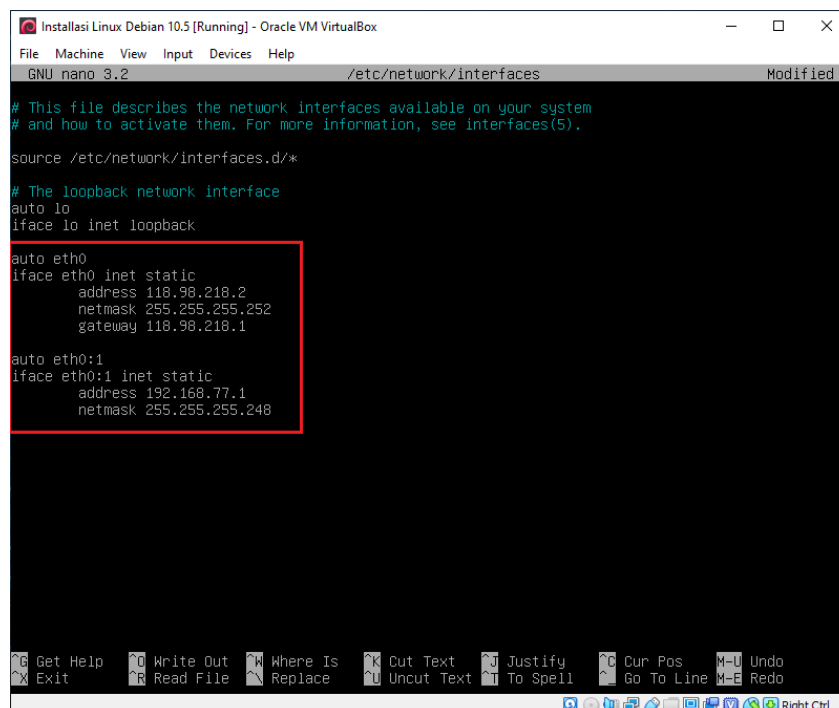
iface eth0 inet static
    address 118.98.218.2
    netmask 255.255.255.252
    gateway 118.98.218.1
```

→ (IP WAN)
→ (Netmask)
→ (IP Modem)

```
auto eth0:1

iface eth0:1 inet static
    address 192.168.77.1
    netmask 255.255.255.248
```

→ (IP LAN)
→ (Netmask)



```
Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
GNU nano 3.2 /etc/network/interfaces Modified

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

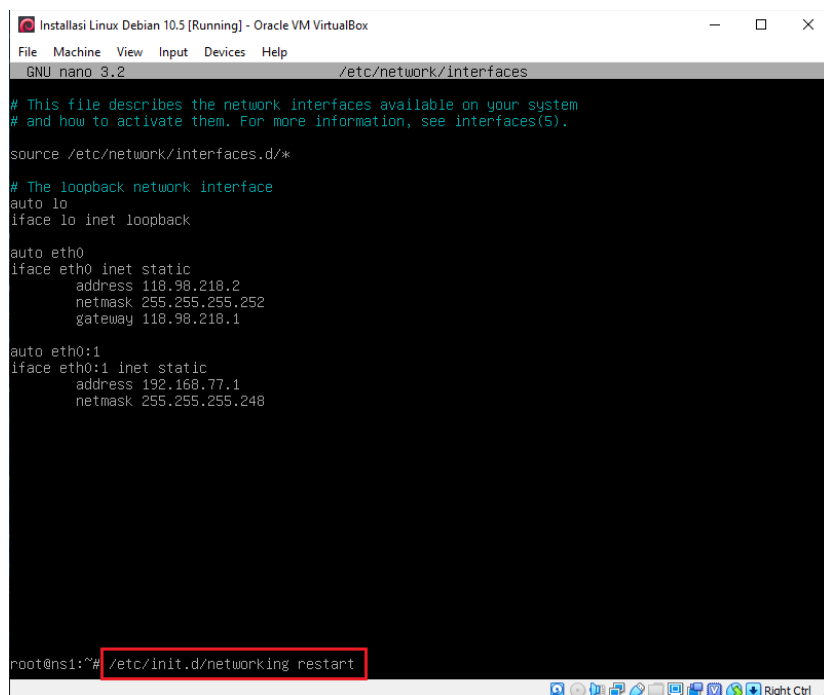
auto eth0
iface eth0 inet static
    address 118.98.218.2
    netmask 255.255.255.252
    gateway 118.98.218.1

auto eth0:1
iface eth0:1 inet static
    address 192.168.77.1
    netmask 255.255.255.248
```

15. Lalu save dengan cara menekan tombol kombinasi **Ctrl + X** lalu tekan tombol **Y** kemudian **Enter**.

16. Masukkan perintah berikut untuk merestart network-nya :

/etc/init.d/networking restart >> lalu tekan Enter



```
GNU nano 3.2 /etc/network/interfaces

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

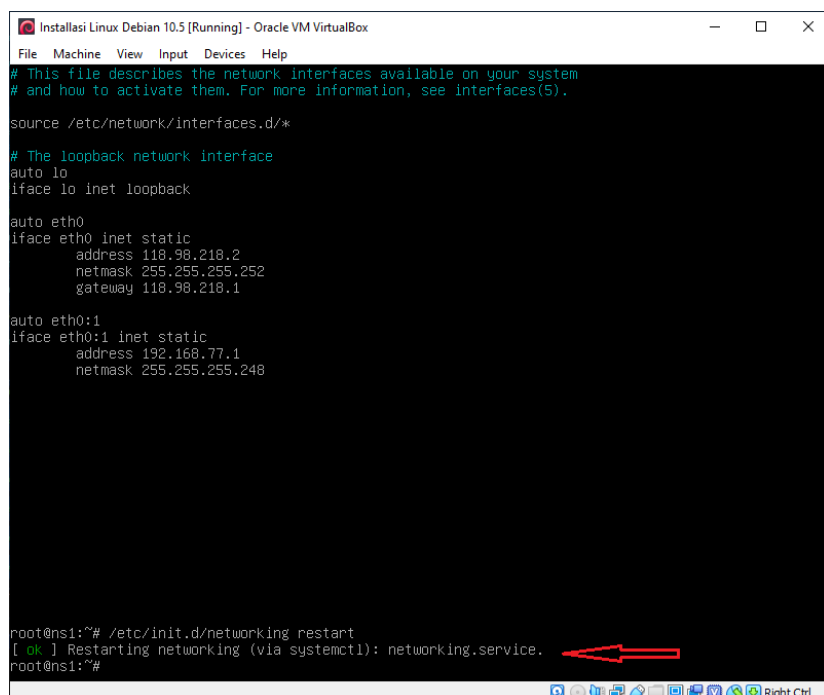
# The loopback network interface
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet static
    address 118.98.218.2
    netmask 255.255.255.252
    gateway 118.98.218.1

auto eth0:1
iface eth0:1 inet static
    address 192.168.77.1
    netmask 255.255.255.248

root@ns1:~# /etc/init.d/networking restart
```

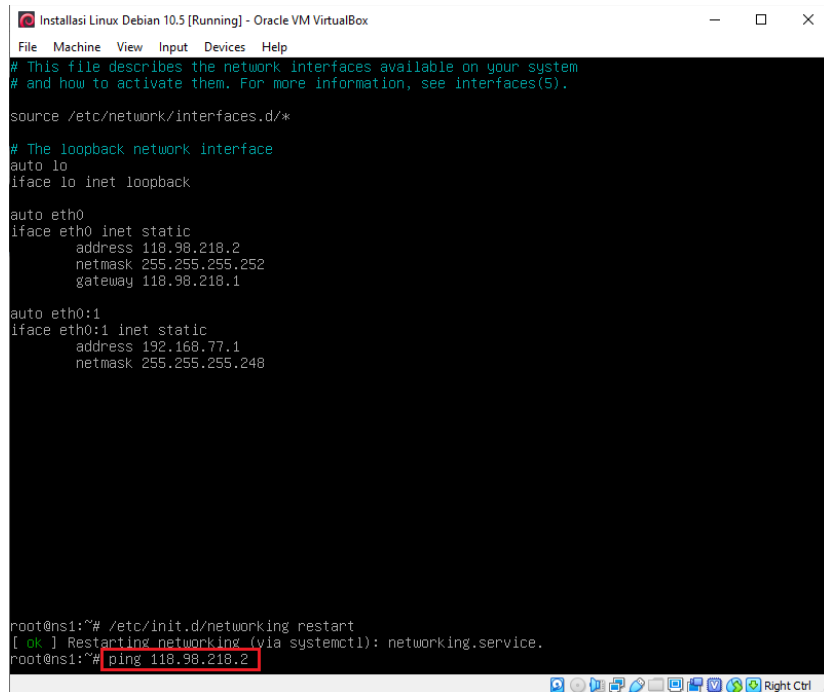
17. Pastikan hasilnya (**OK**) seperti gambar di bawah :



```
root@ns1:~# /etc/init.d/networking restart
[ OK ] Restarting networking (via systemctl): networking.service.
root@ns1:~#
```

18. Lakukan tes ping pada IP WAN yang telah di daftarkan tadi, dengan cara masukkan perintah berikut :

```
# ping 118.98.218.2
```



```
Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

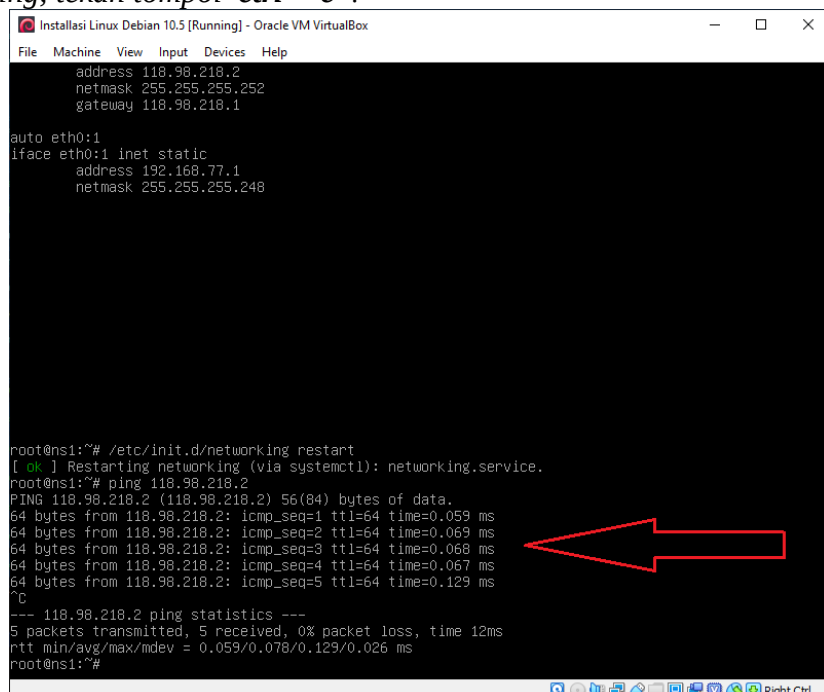
# The loopback network interface
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet static
    address 118.98.218.2
    netmask 255.255.255.252
    gateway 118.98.218.1

auto eth0:1
iface eth0:1 inet static
    address 192.168.77.1
    netmask 255.255.255.248

root@ns1:~# /etc/init.d/networking restart
[ OK ] Restarting networking (via systemctl): networking.service.
root@ns1:~# ping 118.98.218.2
```

19. Jika berhasil maka hasilnya seperti pada gambar di bawah. “untuk menghentikan proses ping, tekan tombol **ctrl + c**”.



```
Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

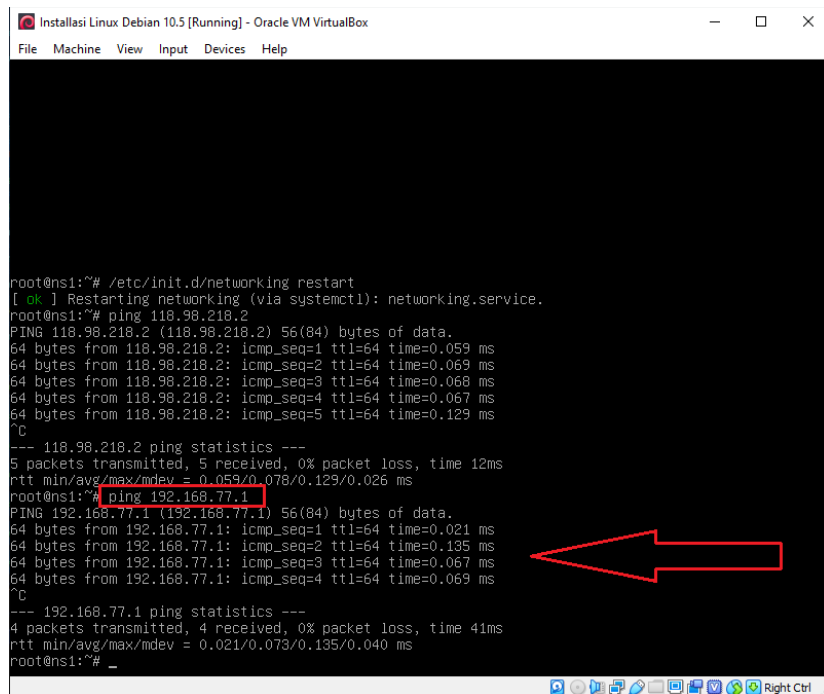
address 118.98.218.2
netmask 255.255.255.252
gateway 118.98.218.1

auto eth0:1
iface eth0:1 inet static
    address 192.168.77.1
    netmask 255.255.255.248

root@ns1:~# /etc/init.d/networking restart
[ OK ] Restarting networking (via systemctl): networking.service.
root@ns1:~# ping 118.98.218.2
PING 118.98.218.2 (118.98.218.2) 56(84) bytes of data:
64 bytes from 118.98.218.2: icmp_seq=1 ttl=64 time=0.059 ms
64 bytes from 118.98.218.2: icmp_seq=2 ttl=64 time=0.069 ms
64 bytes from 118.98.218.2: icmp_seq=3 ttl=64 time=0.068 ms
64 bytes from 118.98.218.2: icmp_seq=4 ttl=64 time=0.067 ms
64 bytes from 118.98.218.2: icmp_seq=5 ttl=64 time=0.129 ms
^C
--- 118.98.218.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 12ms
rtt min/avg/max/mdev = 0.059/0.078/0.129/0.026 ms
root@ns1:~#
```

20. Lakukan lagi tes ping pada IP LAN yang telah di daftarkan tadi, dengan cara masukkan perintah berikut :

ping 192.168.77.1

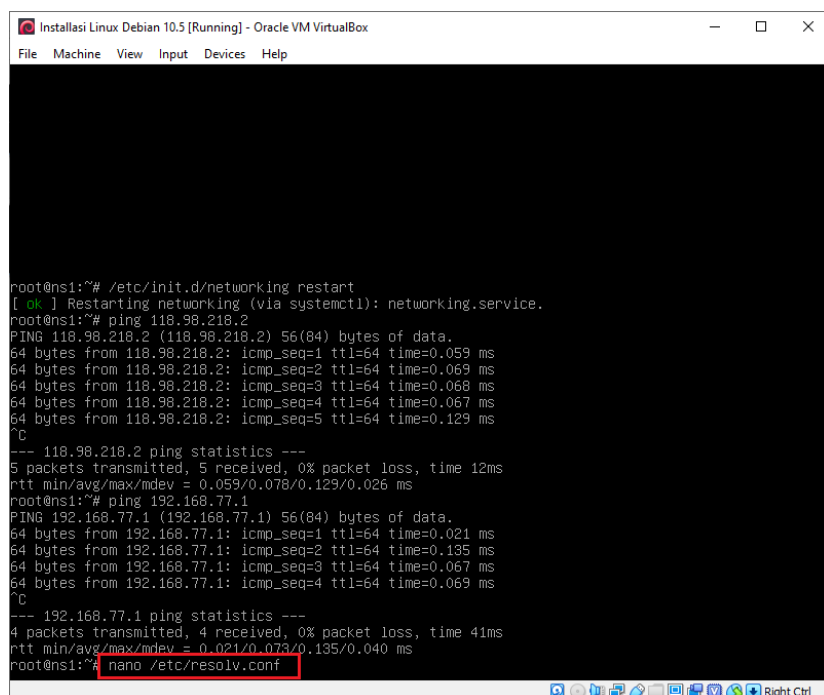


```
root@ns1:~# /etc/init.d/networking restart
[ OK ] Restarting networking (via systemctl): networking.service.
root@ns1:~# ping 118.98.218.2
PING 118.98.218.2 (118.98.218.2) 56(84) bytes of data.
64 bytes from 118.98.218.2: icmp_seq=1 ttl=64 time=0.059 ms
64 bytes from 118.98.218.2: icmp_seq=2 ttl=64 time=0.069 ms
64 bytes from 118.98.218.2: icmp_seq=3 ttl=64 time=0.068 ms
64 bytes from 118.98.218.2: icmp_seq=4 ttl=64 time=0.067 ms
64 bytes from 118.98.218.2: icmp_seq=5 ttl=64 time=0.129 ms
^C
--- 118.98.218.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 12ms
rtt min/avg/max/mdev = 0.059/0.078/0.129/0.026 ms
root@ns1:~# ping 192.168.77.1
PING 192.168.77.1 (192.168.77.1) 56(84) bytes of data.
64 bytes from 192.168.77.1: icmp_seq=1 ttl=64 time=0.021 ms
64 bytes from 192.168.77.1: icmp_seq=2 ttl=64 time=0.135 ms
64 bytes from 192.168.77.1: icmp_seq=3 ttl=64 time=0.067 ms
64 bytes from 192.168.77.1: icmp_seq=4 ttl=64 time=0.069 ms
^C
--- 192.168.77.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 41ms
rtt min/avg/max/mdev = 0.021/0.073/0.135/0.040 ms
root@ns1:~#
```

21. Jika berhasil maka hasilnya seperti pada gambar di atas. “untuk menghentikan proses ping, tekan tombol **ctrl + c**”.

22. Langkah selanjutnya yaitu daftarkan IP DNS dengan cara ketikkan perintah berikut :

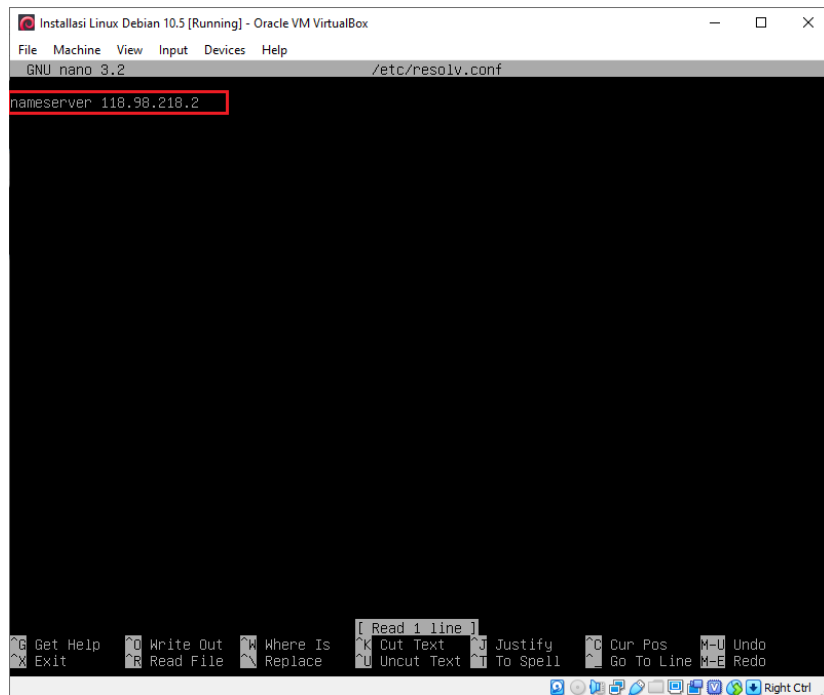
nano /etc/resolv.conf >> lalu tekan Enter



```
root@ns1:~# /etc/init.d/networking restart
[ OK ] Restarting networking (via systemctl): networking.service.
root@ns1:~# ping 118.98.218.2
PING 118.98.218.2 (118.98.218.2) 56(84) bytes of data.
64 bytes from 118.98.218.2: icmp_seq=1 ttl=64 time=0.059 ms
64 bytes from 118.98.218.2: icmp_seq=2 ttl=64 time=0.069 ms
64 bytes from 118.98.218.2: icmp_seq=3 ttl=64 time=0.068 ms
64 bytes from 118.98.218.2: icmp_seq=4 ttl=64 time=0.067 ms
64 bytes from 118.98.218.2: icmp_seq=5 ttl=64 time=0.129 ms
^C
--- 118.98.218.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 12ms
rtt min/avg/max/mdev = 0.059/0.078/0.129/0.026 ms
root@ns1:~# ping 192.168.77.1
PING 192.168.77.1 (192.168.77.1) 56(84) bytes of data.
64 bytes from 192.168.77.1: icmp_seq=1 ttl=64 time=0.021 ms
64 bytes from 192.168.77.1: icmp_seq=2 ttl=64 time=0.135 ms
64 bytes from 192.168.77.1: icmp_seq=3 ttl=64 time=0.067 ms
64 bytes from 192.168.77.1: icmp_seq=4 ttl=64 time=0.069 ms
^C
--- 192.168.77.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 41ms
rtt min/avg/max/mdev = 0.021/0.073/0.135/0.040 ms
root@ns1:~# nano /etc/resolv.conf
```

23. Ketikkan script berikut :

```
nameserver 118.98.218.2
```

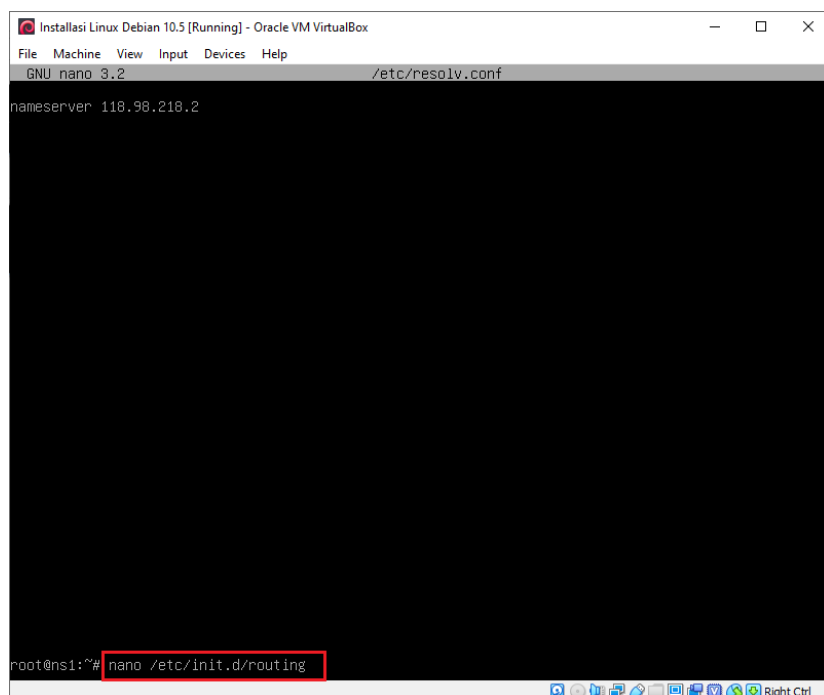


```
Installasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
GNU nano 3.2 /etc/resolv.conf
nameserver 118.98.218.2
```

24. Lalu save dengan cara menekan tombol kombinasi **Ctrl + X** lalu tekan tombol **Y** kemudian **Enter**.

25. Langkah selanjutnya yaitu membuat tabel routingnya. Ketikkan perintah berikut :

```
# nano /etc/init.d/routing >> lalu tekan Enter
```



```
Installasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
GNU nano 3.2 /etc/resolv.conf
nameserver 118.98.218.2

root@ns1:~# nano /etc/init.d/routing
```

26. Ketikkan script berikut :

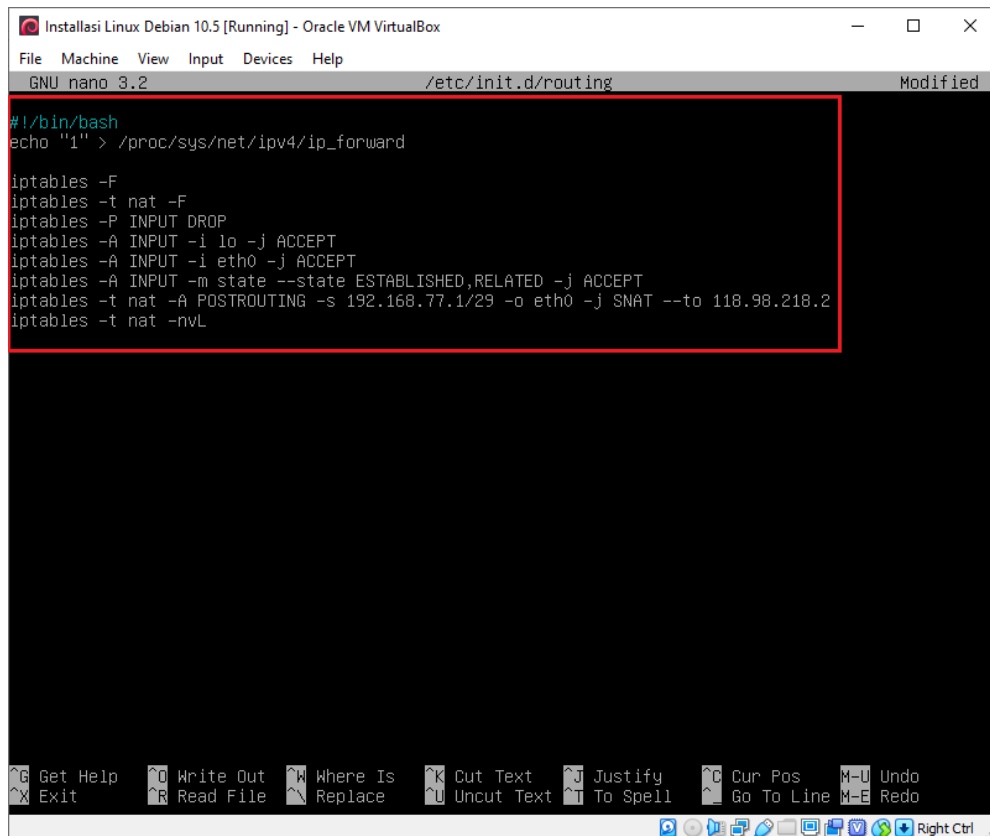
```
#!/bin/bash

echo "1" > /proc/sys/net/ipv4/ip_forward
iptables -F

iptables -t nat -F
iptables -P INPUT DROP

iptables -A INPUT -i lo -j ACCEPT
iptables -A INPUT -i eth0 -j ACCEPT

iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT
iptables -t nat -A POSTROUTING -s 192.168.77.1/29 -o eth0 -j SNAT --to 118.98.218.2
iptables -t nat -nvL
```



```
Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
GNU nano 3.2 /etc/init.d/routing Modified

#!/bin/bash
echo "1" > /proc/sys/net/ipv4/ip_forward

iptables -F
iptables -t nat -F
iptables -P INPUT DROP
iptables -A INPUT -i lo -j ACCEPT
iptables -A INPUT -i eth0 -j ACCEPT
iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT
iptables -t nat -A POSTROUTING -s 192.168.77.1/29 -o eth0 -j SNAT --to 118.98.218.2
iptables -t nat -nvL

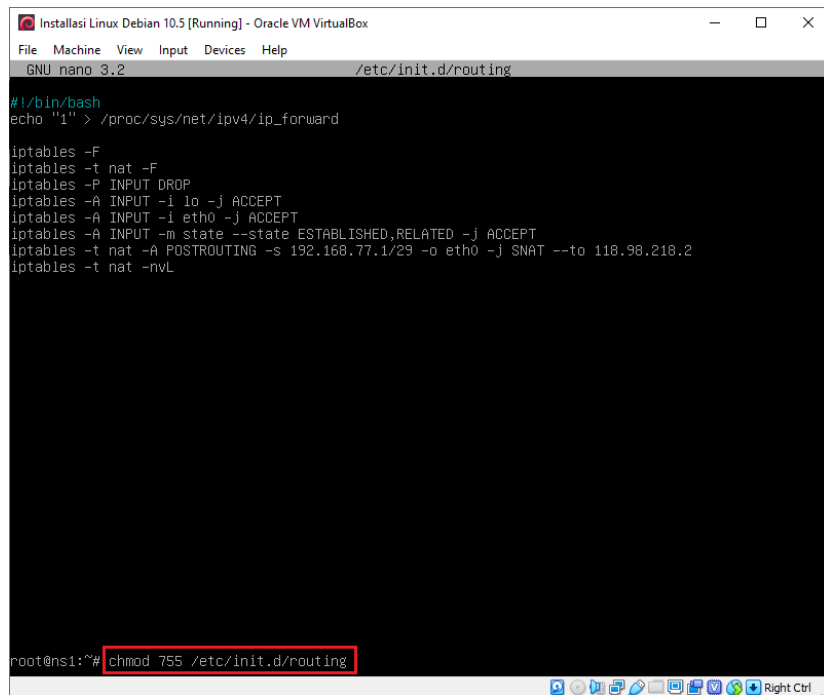
^G Get Help  ^O Write Out  ^W Where Is  ^K Cut Text   ^J Justify    ^C Cur Pos   M-U Undo
^X Exit      ^R Read File  ^N Replace   ^U Uncut Text ^I To Spell   ^_ Go To Line  M-E Redo
Right Ctrl
```

27. Lalu save dengan cara menekan tombol kombinasi **Ctrl + X** lalu tekan tombol **Y** kemudian **Enter**.

28. Langkah selanjutnya yaitu memberikan izin akses untuk file routing yang tadi di buat.
Ketikkan perintah berikut :

chmod 755 /etc/init.d/routing

>> lalu tekan Enter



```
Installasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
GNU nano 3.2 /etc/init.d/routing

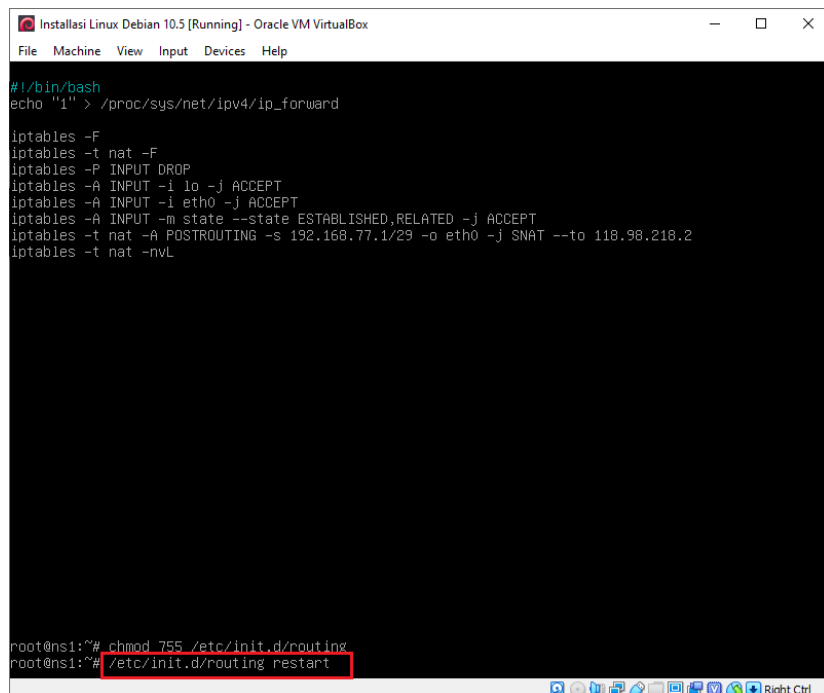
#!/bin/bash
echo "1" > /proc/sys/net/ipv4/ip_forward

iptables -F
iptables -t nat -F
iptables -P INPUT DROP
iptables -A INPUT -i lo -j ACCEPT
iptables -A INPUT -i eth0 -j ACCEPT
iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT
iptables -t nat -A POSTROUTING -s 192.168.77.1/29 -o eth0 -j SNAT --to 118.98.218.2
iptables -t nat -nvL

root@ns1:~# chmod 755 /etc/init.d/routing
```

29. Kemudian restart routingnya. Ketikkan perintah berikut :

/etc/init.d/routing restart *>> lalu tekan Enter*



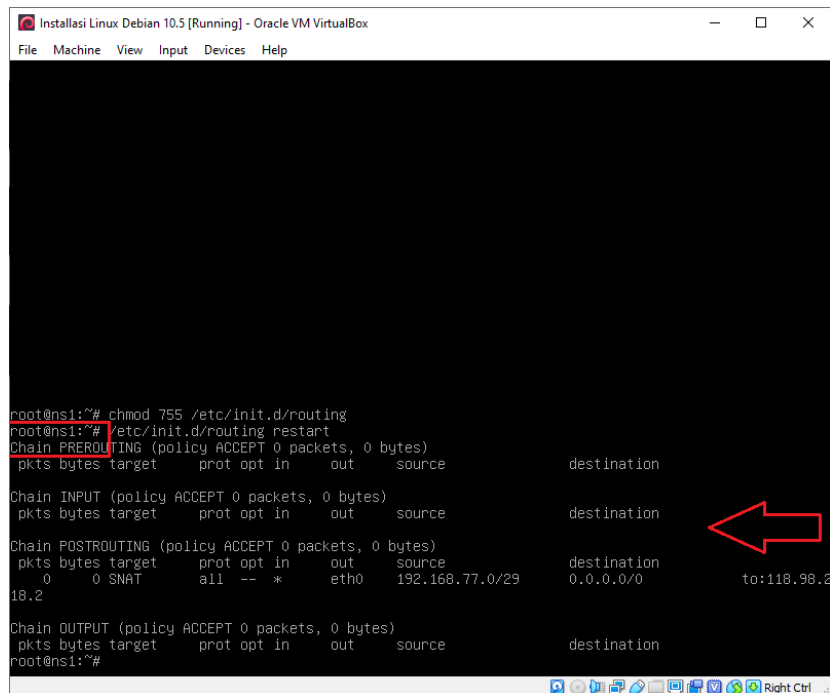
```
Installasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

#!/bin/bash
echo "1" > /proc/sys/net/ipv4/ip_forward

iptables -F
iptables -t nat -F
iptables -P INPUT DROP
iptables -A INPUT -i lo -j ACCEPT
iptables -A INPUT -i eth0 -j ACCEPT
iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT
iptables -t nat -A POSTROUTING -s 192.168.77.1/29 -o eth0 -j SNAT --to 118.98.218.2
iptables -t nat -nvL

root@ns1:~# chmod 755 /etc/init.d/routing
root@ns1:~# /etc/init.d/routing restart
```

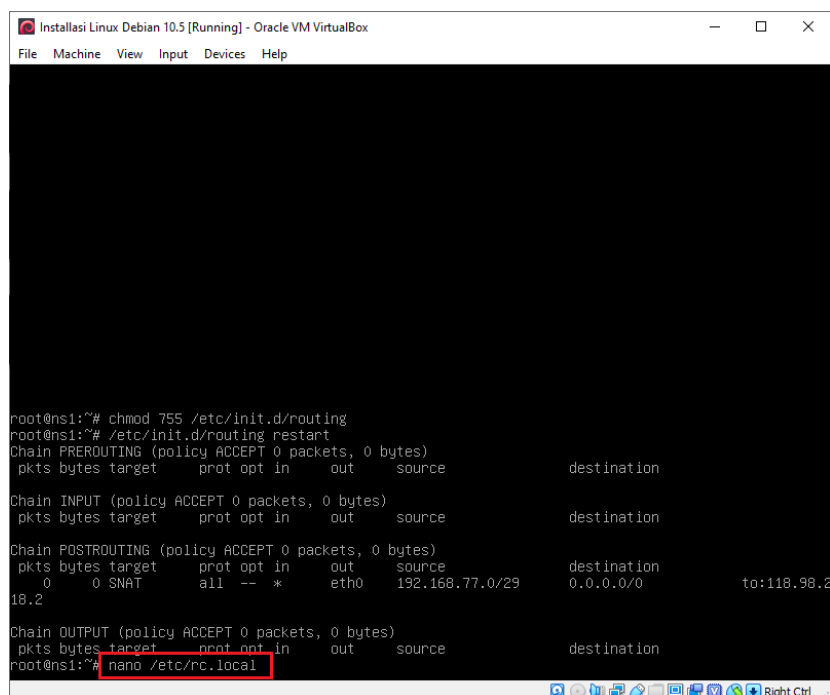
30. Restart routingnya berhasil apabila tidak ada pesan error yang di tampilkan seperti pada gambar di bawah. “Jika ada pesan error maka akan ditampilak di antara **root@ns1:~#** dan **Chain**”.



```
root@ns1:~# chmod 755 /etc/init.d/routing
root@ns1:~# /etc/init.d/routing restart
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source            destination
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source            destination
Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source            destination
    0    0 SNAT      all  --  *      eth0    192.168.77.0/29   0.0.0.0/0         to:118.98.2
18.2
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source            destination
root@ns1:~#
```

31. Langkah selanjutnya yaitu mengaktifkan atau menjalankan secara otomatis script yang ada di dalam file routing yang di buat tadi. Ketikkan perintah berikut :

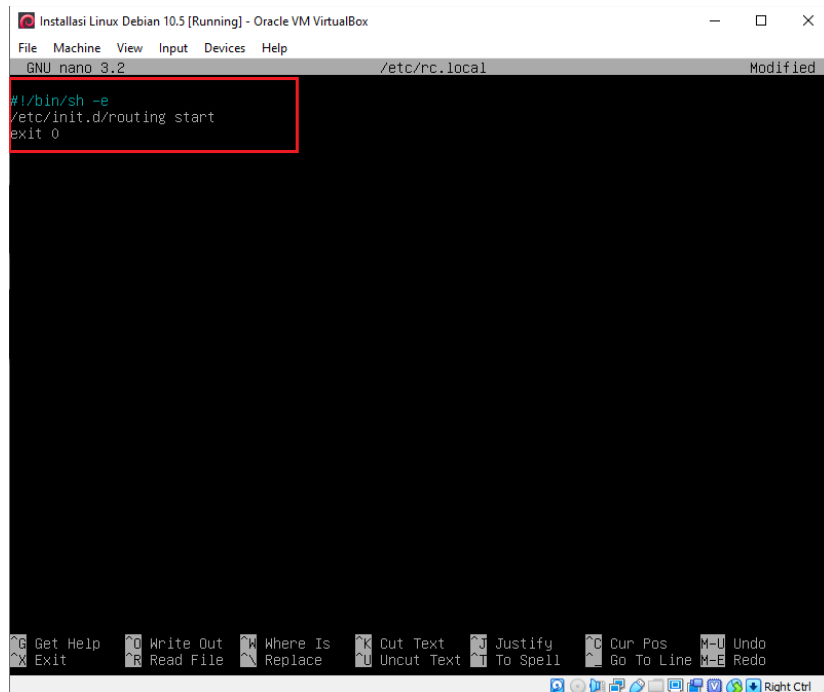
nano /etc/rc.local >> lalu tekan Enter



```
root@ns1:~# chmod 755 /etc/init.d/routing
root@ns1:~# /etc/init.d/routing restart
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source            destination
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source            destination
Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source            destination
    0    0 SNAT      all  --  *      eth0    192.168.77.0/29   0.0.0.0/0         to:118.98.2
18.2
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source            destination
root@ns1:~# nano /etc/rc.local
```

32. Tambahkan script berikut :

```
#!/bin/sh -e
/etc/init.d/routing start
exit 0
```



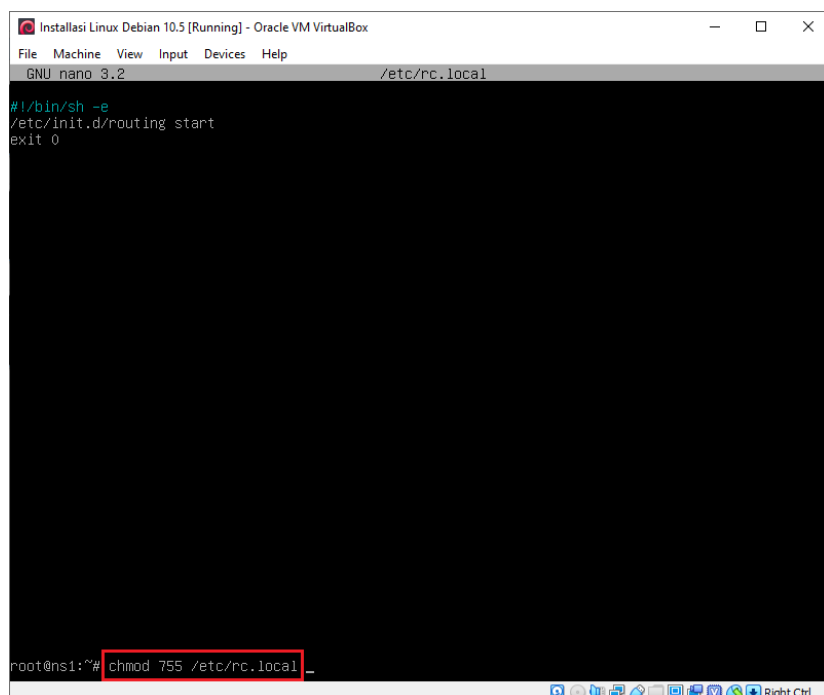
The screenshot shows a terminal window titled "Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox". Inside, the nano 3.2 editor is open, editing the file /etc/rc.local. The text in the editor is: `#!/bin/sh -e`, `/etc/init.d/routing start`, and `exit 0`. This text is enclosed in a red rectangular box. The nano editor's status bar at the bottom shows various keyboard shortcuts like "Get Help", "Write Out", "Where Is", "Cut Text", "Justify", "Cur Pos", "Undo", "Exit", "Read File", "Replace", "Uncut Text", "To Spell", "Go To Line", "Redo", and "Right Ctrl".

33. Lalu save dengan cara menekan tombol kombinasi **Ctrl + X** lalu tekan tombol **Y** kemudian **Enter**.

34. Langkah selanjutnya yaitu memberikan izin akses untuk file **rc.local** yang tadi di buat. Ketikkan perintah berikut :

```
# chmod 755 /etc/rc.local
```

>> lalu tekan *Enter*

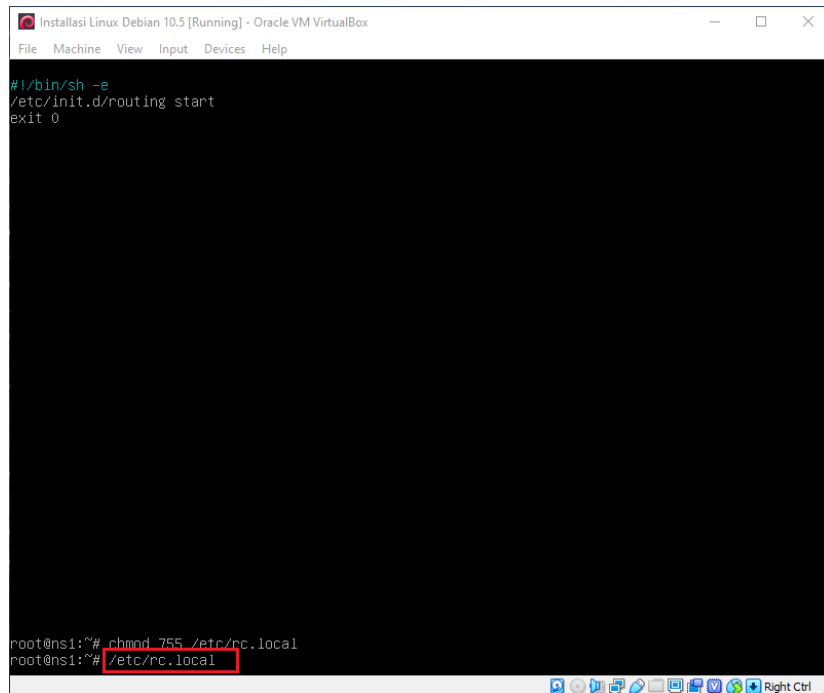


The screenshot shows the same terminal window. The nano editor is still open, showing the same script. Below the editor, in the terminal prompt area, the command `root@ns1:~# chmod 755 /etc/rc.local` has been entered. This command is highlighted with a red rectangular box. The terminal window title and the nano editor interface remain the same.

35. Untuk memastikan script yang kita buat di file **rc.local** sudah berhasil atau belum. Ketikkan perintah berikut :

/etc/rc.local

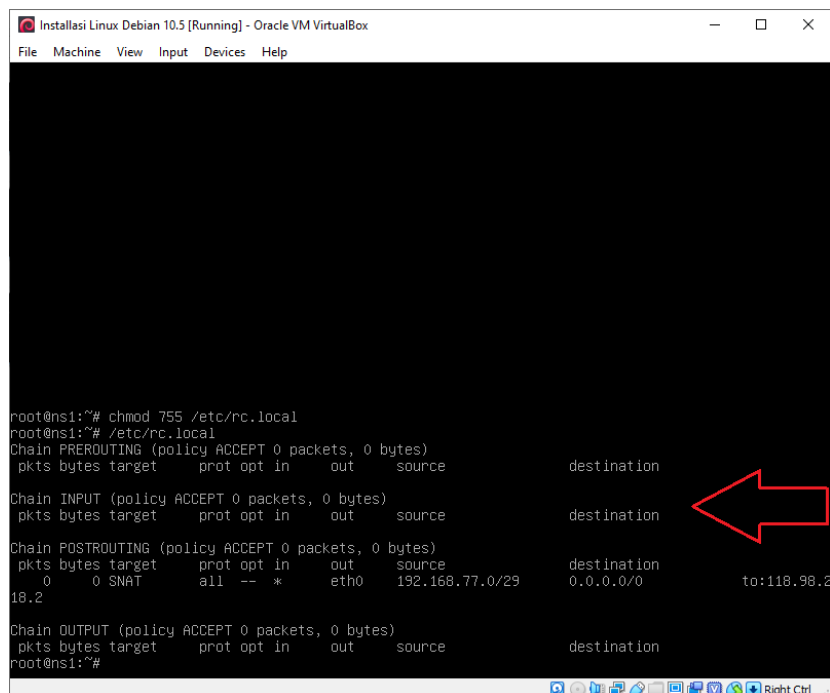
>> *lalu tekan Enter*



```
#!/bin/sh -e
/etc/init.d/routing start
exit 0

root@ns1:~# chmod 755 /etc/rc.local
root@ns1:~# /etc/rc.local
```

36. Jika berhasil maka tampilannya sama ketika melakukan restart routing seperti gambar di bawah.



```
root@ns1:~# chmod 755 /etc/rc.local
root@ns1:~# /etc/rc.local
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source    destination
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source    destination
Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source    destination
    0      0 SNAT      all  --  *      etho    192.168.77.0/29    0.0.0.0/0    to:118.98.2
18.2
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source    destination
root@ns1:~#
```

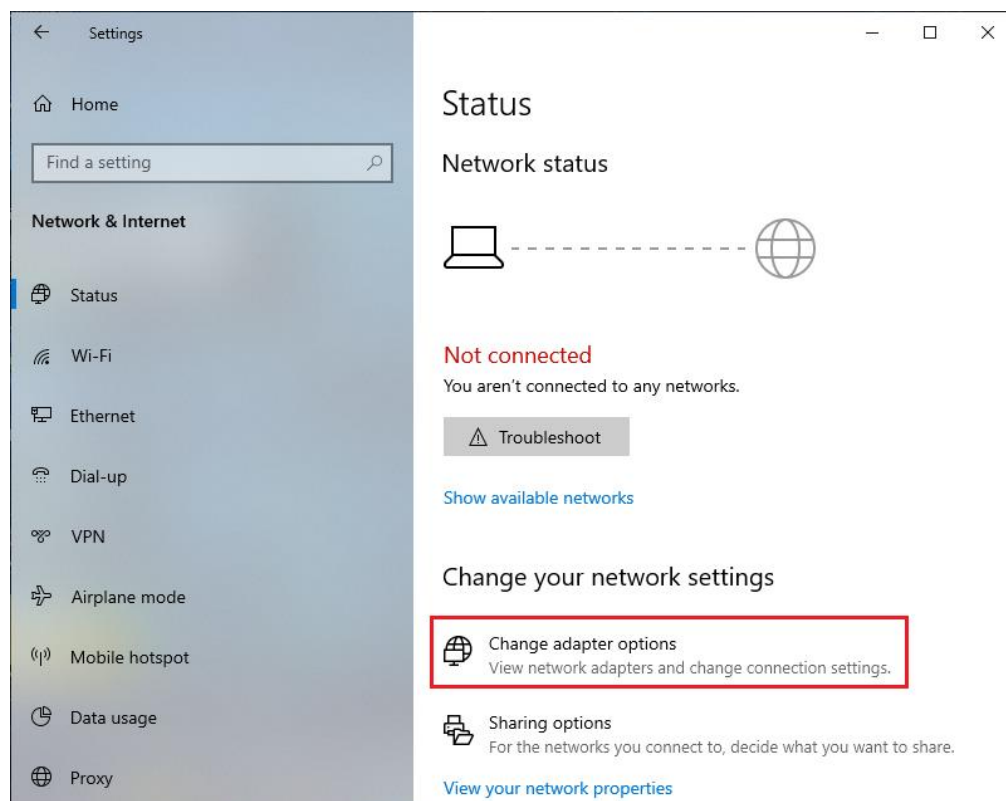
37. Sampai pada step ini, konfigurasi Router Server telah selesai. Selanjutnya melakukan setting IP atau pendaftaran IP pada Client (Windows)
38. Klik kanan pada ikon jaringan yang berada pada sudut kanan bawah taskbar Windows.



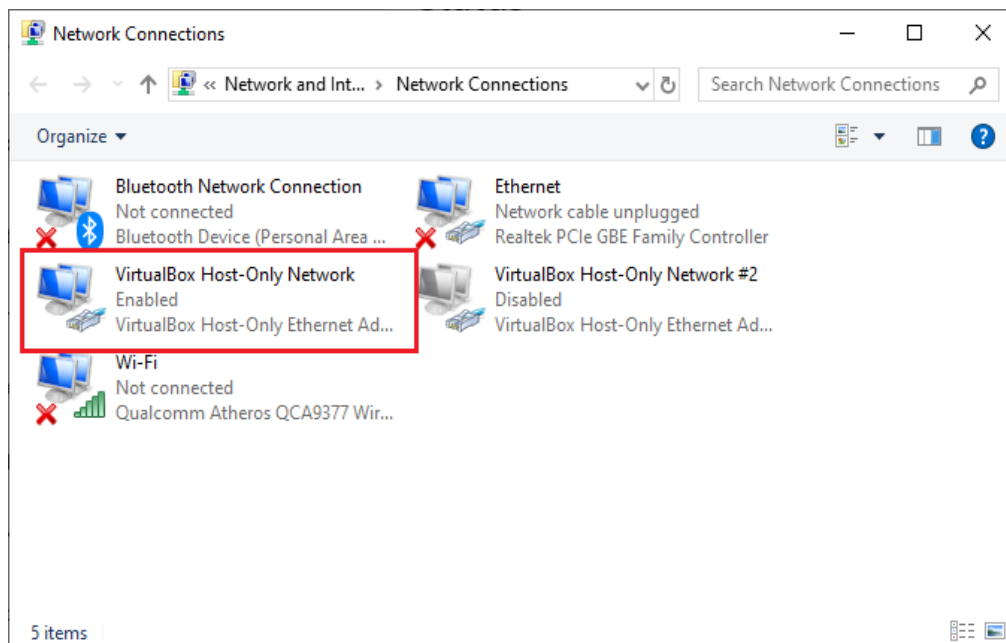
39. Lalu pilih **Open Network & Internet settings**.



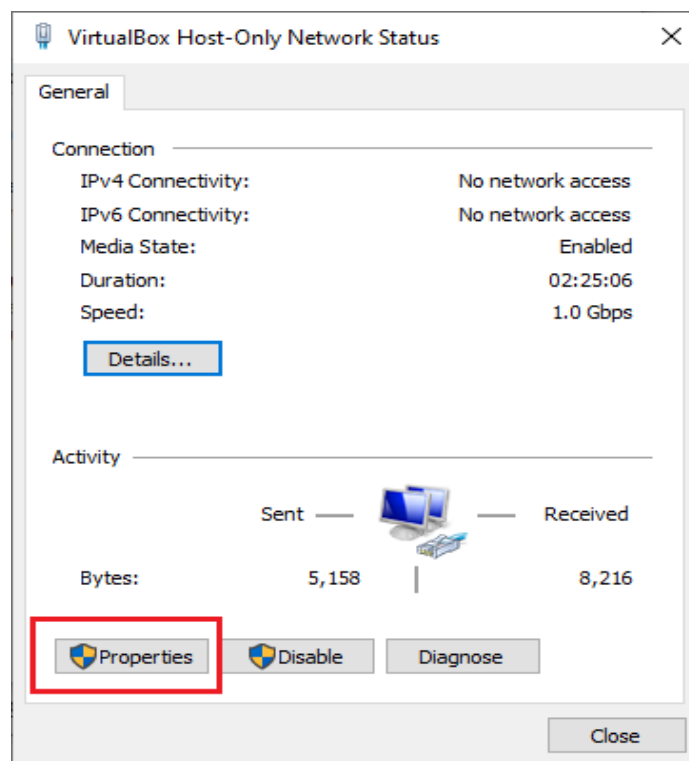
40. Klik **Change adapter options**



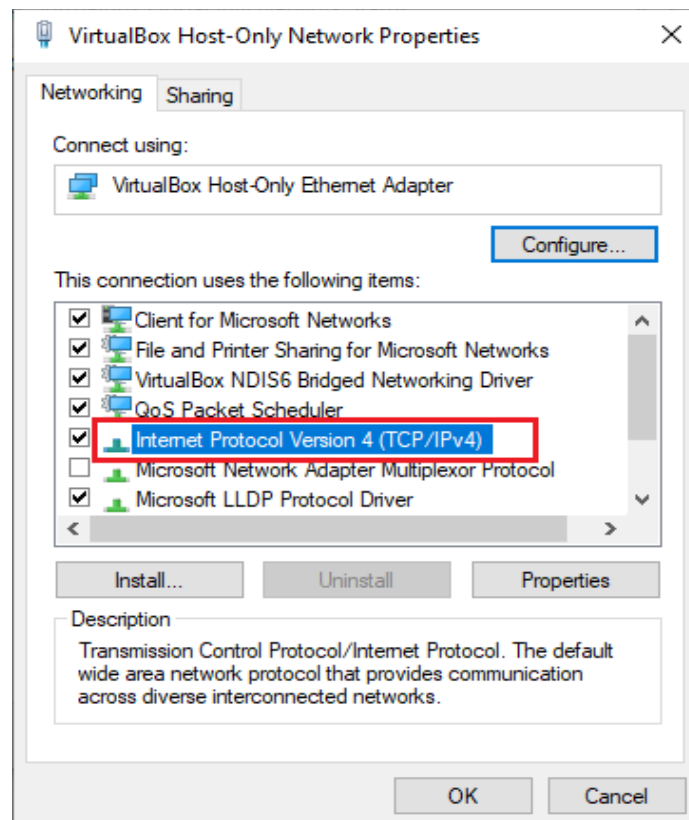
41. Kemudian pilih **VirtualBox Host-Only Network**



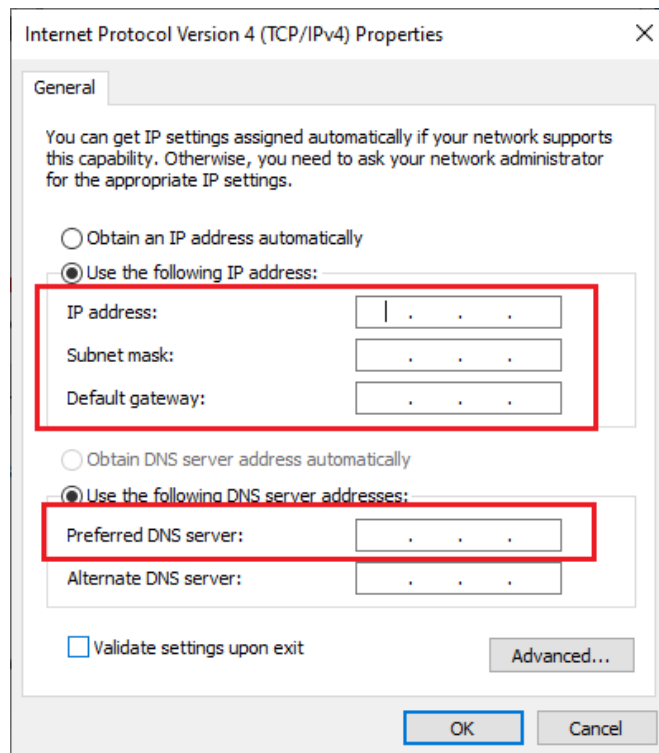
42. Klik **Properties**



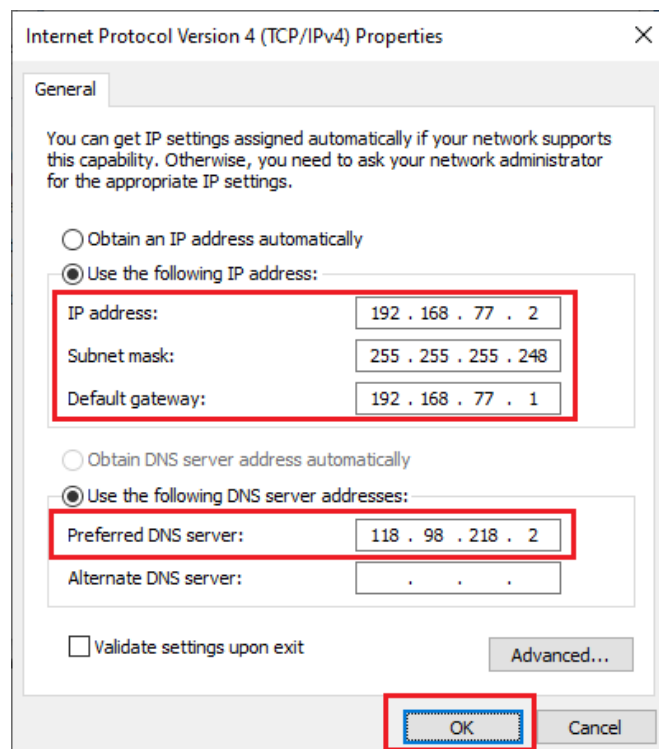
43. Klik **Internet Protocol Version 4(TCP/IPv4)**



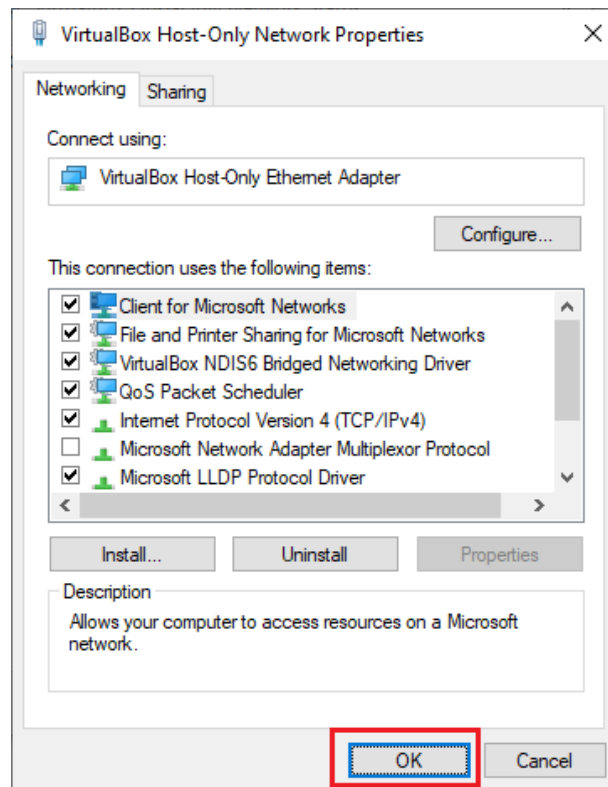
44. Maka akan muncul tampilan seperti berikut. Pilih **Use the following IP address**



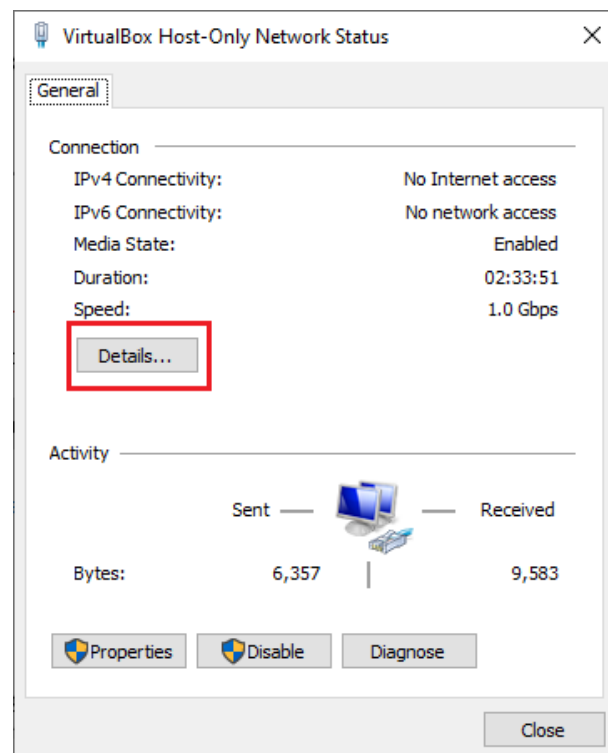
45. Kemudian masukkan alamat IP sesuai dengan Topologi. Contohnya seperti berikut ini. Jika sudah selesai, klik **OK**.



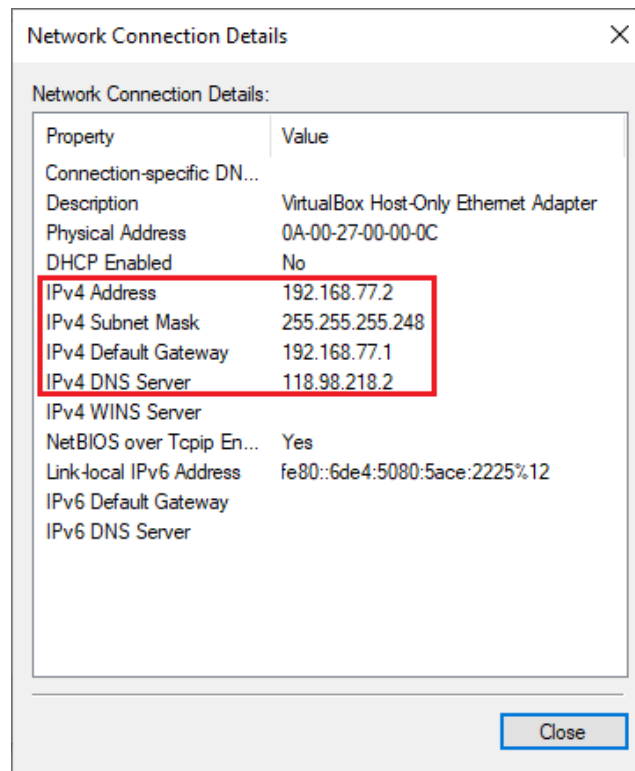
46. Kemudian klik **OK**.



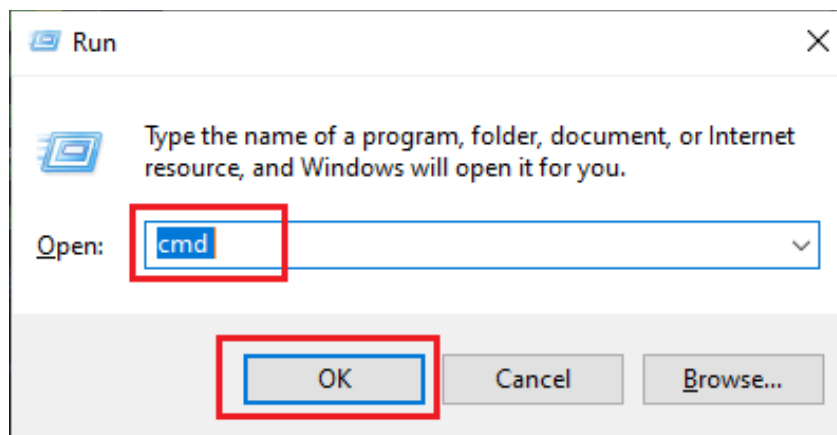
47. Kemudian cek alamat IP yang sudah di daftarkan, dengan cara klik **Details...**



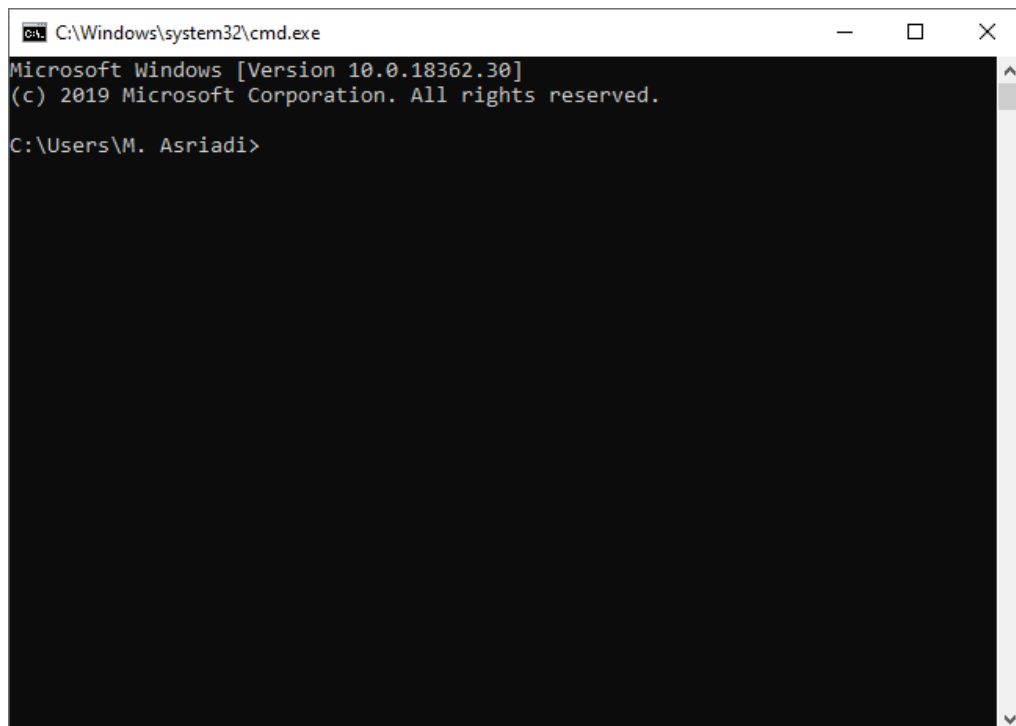
48. Pastikan hasilnya seperti gambar di bawah.



49. Langkah selanjutnya adalah memastikan apakah client sudah terhubung dengan server, dengan cara melakukan tes ping dari Client ke Server. Tekan tombol kombinasi **Windows + R** pada keyboard lalu ketikkan **CMD** kemudian klik **OK** atau tekan Enter.



50. Maka akan tampil terminal seperti di bawah.

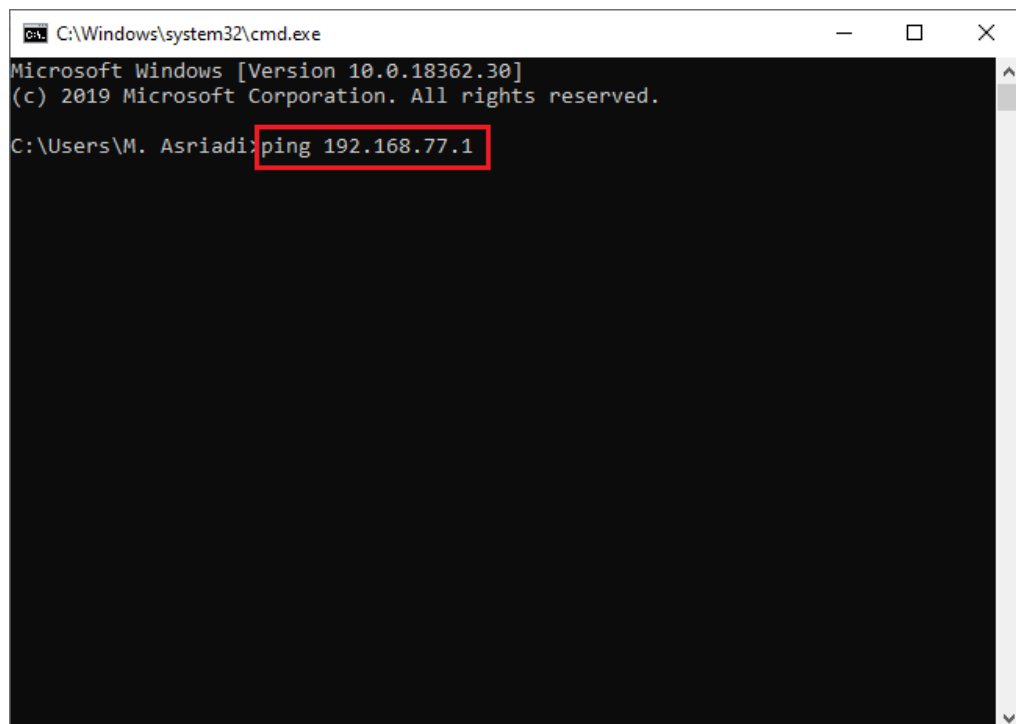


```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.18362.30]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\M. Asriadi>
```

51. Lakukan tes koneksi dengan cara ping IP LAN pada Server seperti berikut :

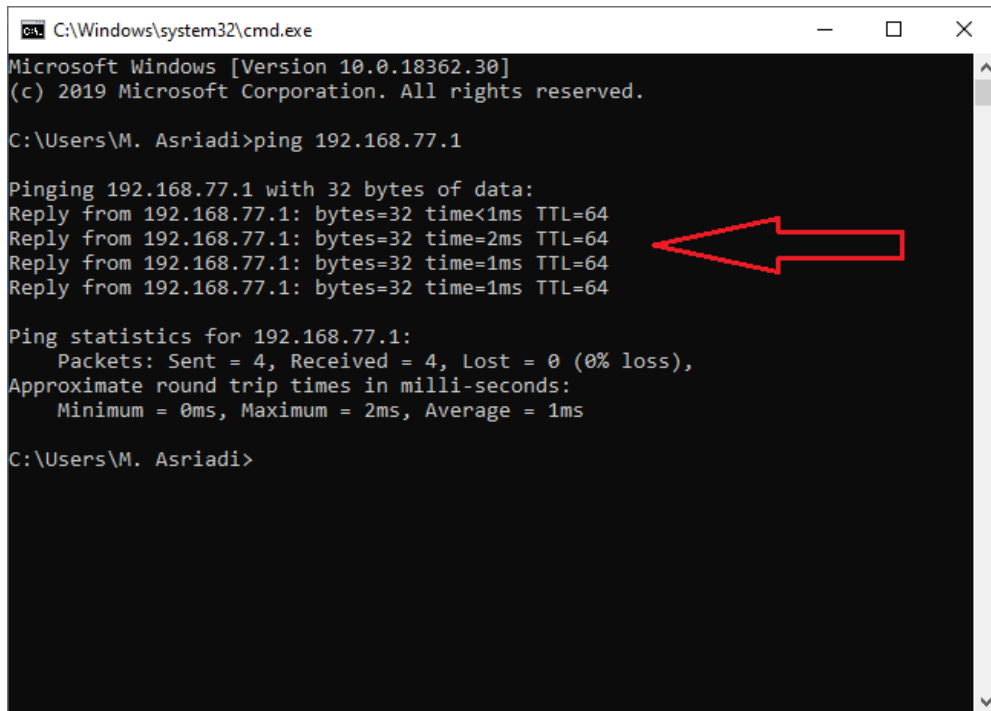
ping 192.168.77.1 >> *lalu tekan Enter*



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.18362.30]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\M. Asriadi>ping 192.168.77.1
```

52. Pastikan hasilnya **Reply** seperti pada gambar di bawah.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.18362.30]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\M. Asriadi>ping 192.168.77.1

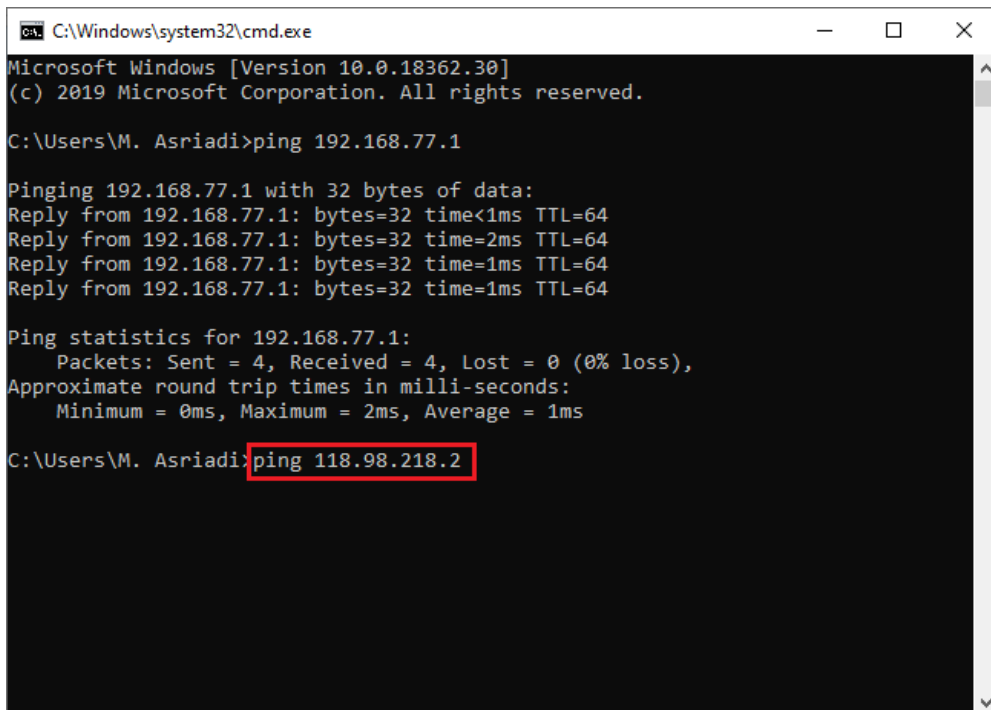
Pinging 192.168.77.1 with 32 bytes of data:
Reply from 192.168.77.1: bytes=32 time<1ms TTL=64
Reply from 192.168.77.1: bytes=32 time=2ms TTL=64
Reply from 192.168.77.1: bytes=32 time=1ms TTL=64
Reply from 192.168.77.1: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.77.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\Users\M. Asriadi>
```

53. Lakukan lagi tes koneksi dengan cara ping IP WAN pada Server seperti berikut :

ping 118.98.218.2 >> lalu tekan Enter



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.18362.30]
(c) 2019 Microsoft Corporation. All rights reserved.

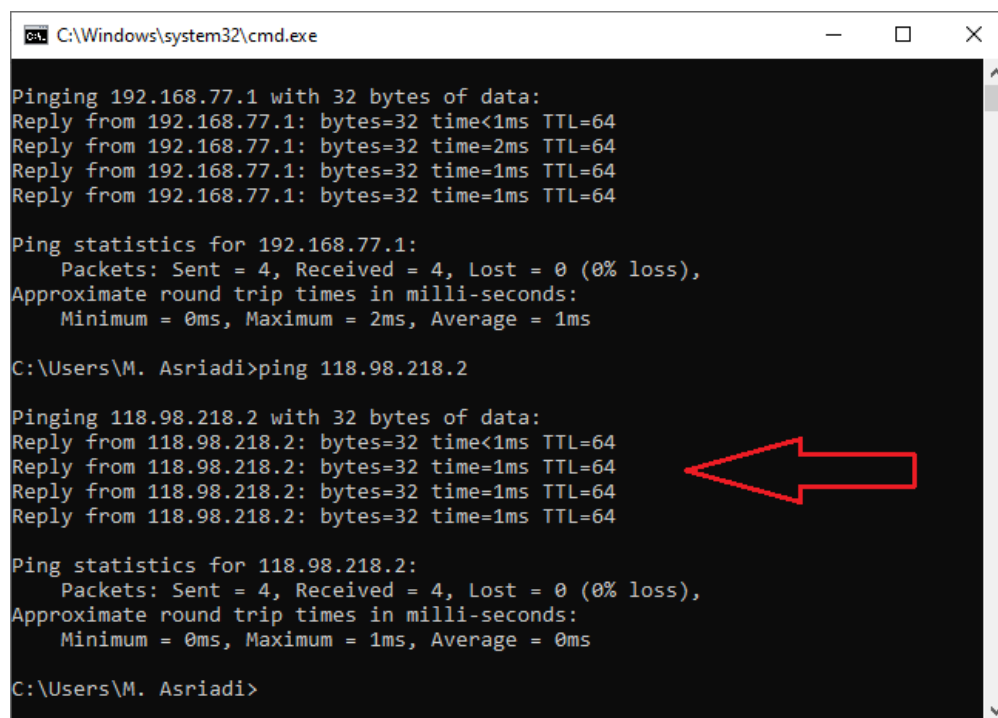
C:\Users\M. Asriadi>ping 192.168.77.1

Pinging 192.168.77.1 with 32 bytes of data:
Reply from 192.168.77.1: bytes=32 time<1ms TTL=64
Reply from 192.168.77.1: bytes=32 time=2ms TTL=64
Reply from 192.168.77.1: bytes=32 time=1ms TTL=64
Reply from 192.168.77.1: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.77.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\Users\M. Asriadi>ping 118.98.218.2
```

54. Pastikan hasilnya **Reply** seperti pada gambar di bawah.



```
C:\Windows\system32\cmd.exe

Pinging 192.168.77.1 with 32 bytes of data:
Reply from 192.168.77.1: bytes=32 time<1ms TTL=64
Reply from 192.168.77.1: bytes=32 time=2ms TTL=64
Reply from 192.168.77.1: bytes=32 time=1ms TTL=64
Reply from 192.168.77.1: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.77.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

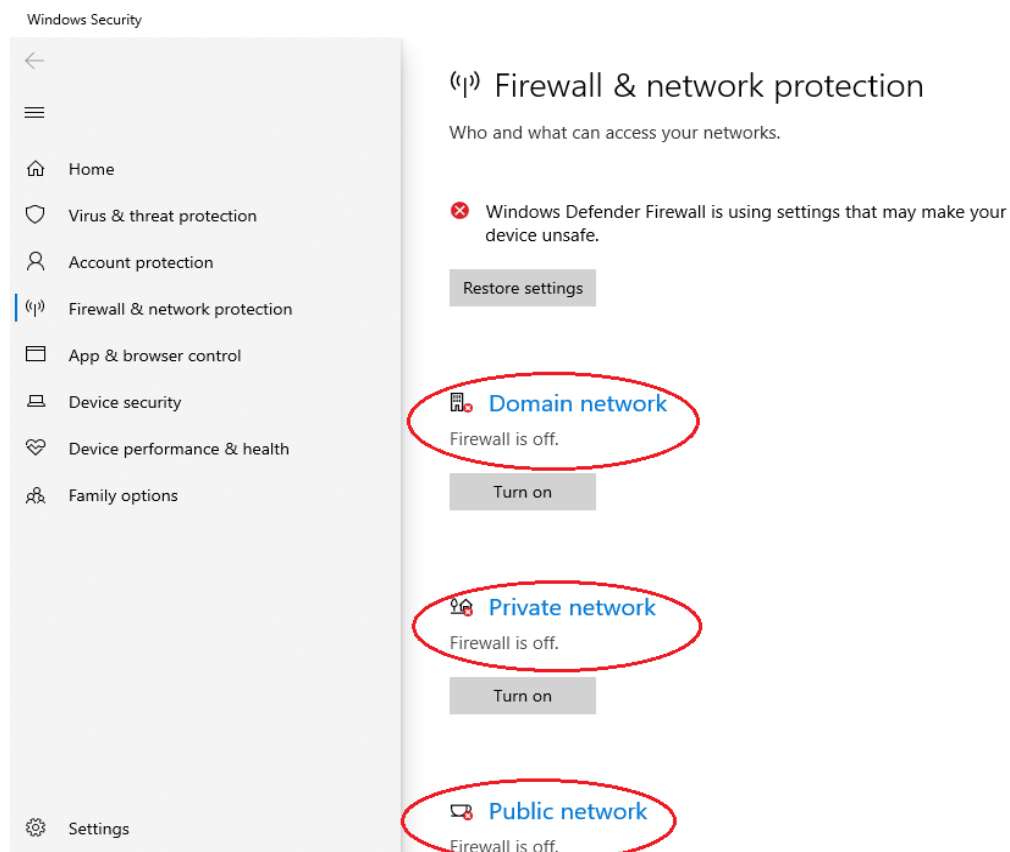
C:\Users\M. Asriadi>ping 118.98.218.2

Pinging 118.98.218.2 with 32 bytes of data:
Reply from 118.98.218.2: bytes=32 time<1ms TTL=64
Reply from 118.98.218.2: bytes=32 time=1ms TTL=64
Reply from 118.98.218.2: bytes=32 time=1ms TTL=64
Reply from 118.98.218.2: bytes=32 time=1ms TTL=64

Ping statistics for 118.98.218.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

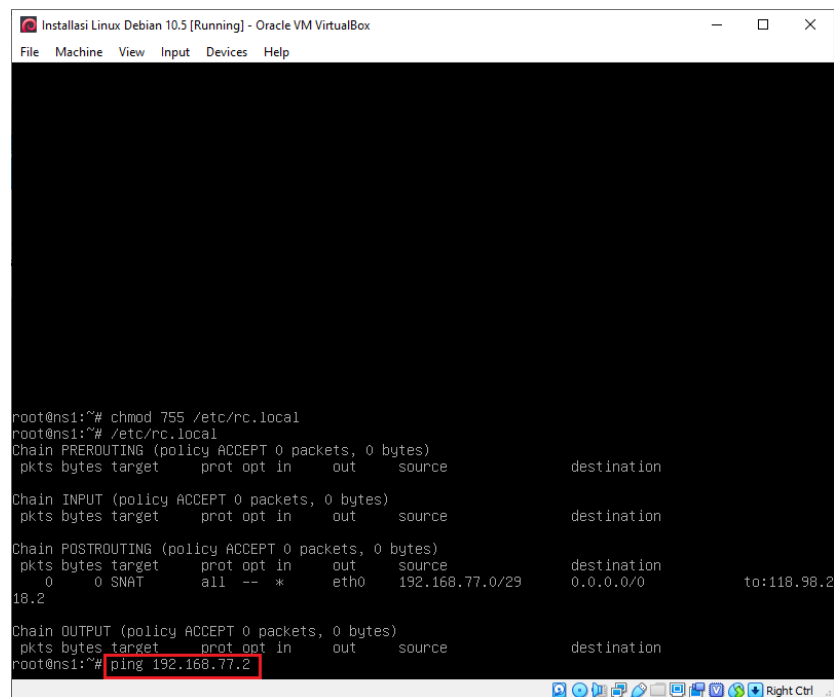
C:\Users\M. Asriadi>
```

55. Langkah terakhir yaitu melakukan tes ping dari Server ke Client. “Sebelum melakukan tes ping dari Server ke Client pastikan terlebih dahulu **Windows Firewall** dalam kondisi tidak aktif (off)”. Serperti gambar berikut :



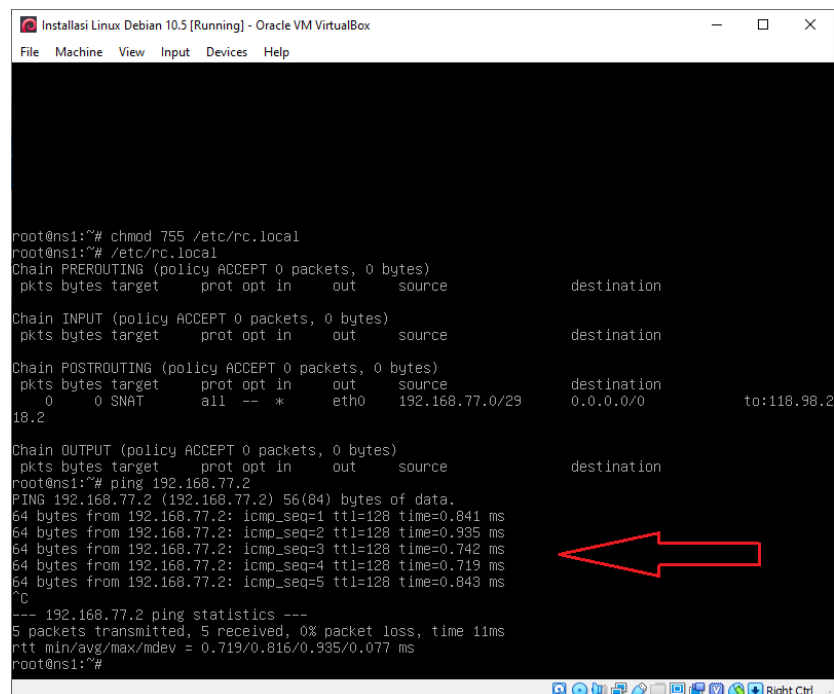
56. Lakukan tes ping dari Server ke Client, dengan cara ping IP Client di Server seperti berikut :

ping 192.168.77.2



```
root@ns1:~# chmod 755 /etc/rc.local
root@ns1:~# /etc/rc.local
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source                   destination
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source                   destination
Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source                   destination
0      0 SNAT     all  --  *      eth0    192.168.77.0/29          0.0.0.0/0               to:118.98.2
18.2
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source                   destination
root@ns1:~# ping 192.168.77.2
```

57. Pastikan hasilnya seperti pada gambar di bawah. “untuk menghentikan proses ping, tekan tombol **ctrl + c**”.



```
root@ns1:~# chmod 755 /etc/rc.local
root@ns1:~# /etc/rc.local
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source                   destination
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source                   destination
Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source                   destination
0      0 SNAT     all  --  *      eth0    192.168.77.0/29          0.0.0.0/0               to:118.98.2
18.2
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source                   destination
root@ns1:~# ping 192.168.77.2
PING 192.168.77.2 (192.168.77.2) 56(84) bytes of data:
64 bytes from 192.168.77.2: icmp_seq=1 ttl=128 time=0.841 ms
64 bytes from 192.168.77.2: icmp_seq=2 ttl=128 time=0.935 ms
64 bytes from 192.168.77.2: icmp_seq=3 ttl=128 time=0.742 ms
64 bytes from 192.168.77.2: icmp_seq=4 ttl=128 time=0.719 ms
64 bytes from 192.168.77.2: icmp_seq=5 ttl=128 time=0.843 ms
^C
--- 192.168.77.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 11ms
rtt min/avg/max/mdev = 0.719/0.816/0.935/0.077 ms
root@ns1:~#
```

58. Jika seluruh step di atas sudah di lakukan, maka konfigurasi Router Server telah selesai dan berhasil.

****SALAM SUKSES TKJ CLUB****